CHAPTER 2

Covered Call Writing

Covered call writing is the name given to the strategy by which one sells a call option while simultaneously owning the obligated number of shares of underlying stock. The writer should be mildly bullish, or at least neutral, toward the underlying stock. By writing a call option against stock, one always decreases the risk of owning the stock. It may even be possible to profit from a covered write if the stock declines somewhat. However, the covered call writer does limit his profit potential and therefore may not fully participate in a strong upward move in the price of the underlying stock. Use of this strategy is becoming so common that the strategist must understand it thoroughly. It is therefore discussed at length.

THE IMPORTANCE OF COVERED CALL WRITING

COVERED CALL WRITING FOR DOWNSIDE PROTECTION

Example: An investor owns 100 shares of XYZ common stock, which is currently selling at $48 per share. If this investor sells an XYZ July 50 call option while still holding his stock, he establishes a covered write. Suppose the investor receives $300 from the sale of the July 50 call. If XYZ is below 50 at July expiration, the call option that was sold expires worthless and the investor earns the $300 that he originally received for writing the call. Thus, he receives $300, or 3 points, of downside protection. That is, he can afford to have the XYZ stock drop by 3 points and still break even on the total transaction. At that time he can write another call option if he so desires.

Note that if the underlying stock should fall by more than 3 points, there will be a loss on the overall position. *Thus, the risk in the covered writing strategy materializes if the stock falls by a distance greater than the call option premium that was originally taken in.*

THE BENEFITS OF AN INCREASE IN STOCK PRICE

If XYZ increases in price moderately, the trader may be able to have the best of both worlds.

Example: If XYZ is at or just below 50 at July expiration, the call still expires worthless, and the investor makes the $300 from the option in addition to having a small profit from his stock purchase. Again, he still owns the stock.

Should XYZ increase in price by expiration to levels above 50, the covered writer has a choice of alternatives. As one alternative, he could do nothing, in which case the option would be assigned and his stock would be called away at the striking price of 50. In that case, his profits would be equal to the $300 received from selling the call plus the profit on the increase of his stock from the purchase price of 48 to the sale price of 50. In this case, however, he would no longer own the stock. If as another alternative he desires to retain his stock ownership, he can elect to buy back (or cover) the written call in the open market. This decision might involve taking a loss on the option part of the covered writing transaction, but he would have a correspondingly larger profit, albeit unrealized, from his stock purchase. Using some specific numbers, one can see how this second alternative works out.

Example: XYZ rises to a price of 60 by July expiration. The call option then sells near its intrinsic value of 10. If the investor covers the call at 10, he loses $700 on the option portion of his covered write. (Recall that he originally received $300 from the sale of the option, and now he is buying it back for $1,000.) However, he relieves the obligation to sell his stock at 50 (the striking price) by buying back the call, so he has an unrealized gain of 12 points in the stock, which was purchased at 48. His total profit, including both realized and unrealized gains, is $500.

This profit is exactly the same as he would have made if he had let his stock be called from him. If called, he would keep the $300 from the sale of the call, and he would make 2 points ($200) from buying the stock at 48 and selling it, via exercise, at 50. This profit, again, is a total of $500. The major difference between the two cases is that the investor no longer owns his stock after letting it be called away, whereas he retains stock ownership if he buys back the written call. Which of the two alternatives is the better one in a given situation is not always clear.

No matter how high the stock climbs in price, the profit from a covered write is limited because the writer has obligated himself to sell stock at the striking price. The covered writer still profits when the stock climbs, but possibly not by as much as he might have had he not written the call. On the other hand, he is receiving $300 of immediate cash inflow, because the writer may take the premium immediately and do with it as he pleases. That income can represent a substantial increase in the income currently provided by the dividends on the underlying stock, or it can act to offset part of the loss in case the stock declines.

For readers who prefer formulae, the profit potential and break-even point of a covered write can be summarized as follows:

Maximum profit potential = Strike price – Stock price + Call price
Downside break-even point = Stock price – Call price

QUANTIFICATION OF THE COVERED WRITE

Table 2-1 and Figure 2-1 depict the *profit graph* for the example involving the XYZ covered write of the July 50 call. The table makes the assumption that the call is bought back at parity. If the stock is called away, the same total profit of $500 results; but the price involved on the stock sale is always 50, and the option profit is always $300.

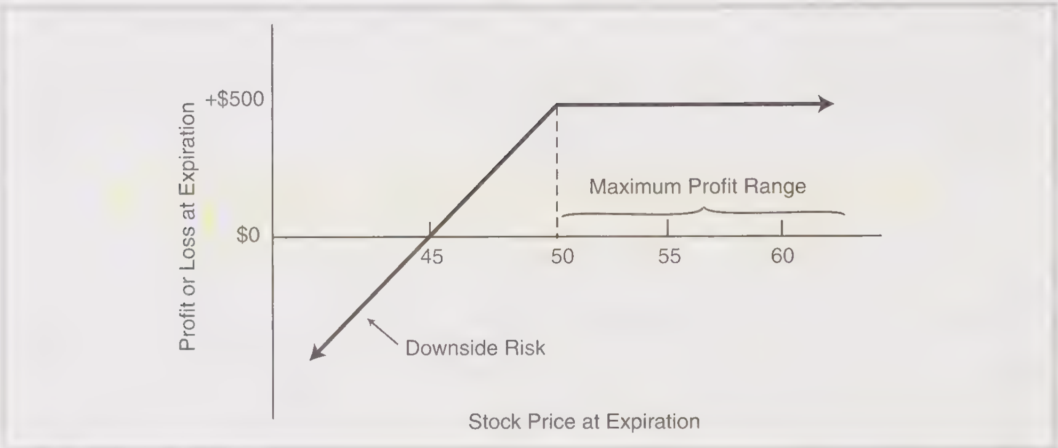
Several conclusions can be drawn. The break-even point is 45 (zero total profit) with risk below 45; the maximum profit attainable is $500 if the position is held until expiration; and the profit if the stock price is unchanged is $300, that is, the covered writer makes $300 even if his stock goes absolutely nowhere.

The profit graph for a covered write always has the shape shown in Figure 2-1. Note that the maximum profit always occurs at all stock prices equal to or greater than the striking price, if the position is held until expiration. However, there is downside risk. If the stock declines in price by too great an amount, the option premium cannot possibly compensate for the entire loss. Downside protective strategies, which are discussed later, attempt to deal with the limitation of this downside risk.

TABLE 2-1.
The XYZ July 50 call.

XYZ Price at Expiration	Stock Profit	July 50 Call at Expiration	Call Profit	Total Profit
40	–$ 800	0	+$300	–$500
45	– 300	0	+ 300	0
48	– 0	0	+ 300	+ 300
50	+ 200	0	+ 300	+ 500
55	+ 700	5	– 200	+ 500
60	+ 1,200	10	– 700	+ 500

FIGURE 2-1.
XYZ covered write.



COVERED WRITING PHILOSOPHY

The primary objective of covered writing, for most investors, is increased income through stock ownership. An ever-increasing number of private and institutional investors are writing call options against the stocks that they own. The facts that the option premium acts as a partial compensation for a decline in price by the underlying stock, and that the premium represents an increase in income to the stockholder, are evident. *The strategy of owning the stock and writing the call will outperform outright stock ownership if the stock falls, remains the same, or even rises slightly.* In fact, the only time that the outright owner of the stock will outperform a covered writer is if the stock increases in price by a relatively substantial amount during the life of the call. Moreover, if one consistently writes call options against his stock, *his portfolio will show less variability of results from quarter to quarter.* The total position—long stock and short option—has less volatility than the stock alone, so on a quarter-by-quarter basis, results will be closer to average than they would be with normal stock ownership. This is an attractive feature, especially for portfolio managers.

However, one should not assume that covered writing will outperform stock ownership. Stocks sometimes tend to make most of their gains in large spurts. A covered writer will not participate in moves such as that. The long-term gains that are quoted for holding stocks include periods of large gains and sometimes periods of large losses as well. The covered writer will not participate in the largest of those gains, since his profit potential is limited.

PHYSICAL LOCATION OF THE STOCK

Before getting more involved in the details of covered writing strategy, it may be useful to review exactly what stock holdings may be written against. Recall that this discussion applies to listed options. If one has deposited stock with his broker in either a cash or a margin account, he may write an option for each 100 shares that he owns without any additional requirement. However, it is possible to write covered options without actually depositing stock with a brokerage firm. There are several ways in which to do this, all involving the deposit of stock with a bank.

Once the stock is deposited with the bank, the investor may have the bank issue an escrow receipt or letter of guarantee to the brokerage firm at which the investor does his option business. The bank must be an "approved" bank in order for the brokerage firm to accept a *letter of guarantee*, and not all firms accept letters of guarantee. These items cost money, and as a new receipt or letter is required for each new option written, the costs may become prohibitive to the customer if only 100 or 200 shares of stock are involved.

There is another alternative open to the customer who wishes to write options without depositing his stock at the brokerage firm. He may deposit his stock with a bank that is a member of the *Depository Trust Corporation* (DTC). The DTC guarantees the Options Clearing Corporation that it will, in fact, deliver stock should an assignment notice be given to the call writer. This is the most convenient method for the investor to use, and is the one used by most of the institutional covered writing investors. There is usually no additional charge for this service by the bank to institutional accounts. However, since only a limited number of banks are members of DTC, and these banks are generally the larger banks located in metropolitan centers, it may be somewhat difficult for many individual investors to take advantage of the DTC opportunity.

TYPES OF COVERED WRITES

While all covered writes involve selling a call against stock that is owned, different terms are used to describe various categories of covered writing. The two broadest terms, under which all covered writes can be classified, are the *out-of-the-money covered write* and the *in-the-money covered write*. These refer, obviously, to whether the option itself was in-the-money or out-of-the-money when the write was first established. Sometimes one may see covered writes classified by the nature of the stock involved (low-priced covered write, high-yield covered write, etc.), but these are only subcases of the two broad categories.

In general, out-of-the-money covered writes offer higher potential rewards but have less risk protection than do in-the-money covered writes. One can establish an aggressive or defensive covered writing position, depending on how far the call option is in- or out-of-the-money when the write is established. In-the-money writes are more defensive covered writing positions.

Some examples may help to illustrate how one covered write can be considerably more conservative, from a strategy viewpoint, than another.

Example: XYZ common stock is selling at 45 and two options are being considered for writing: an XYZ July 40 selling for 8, and an XYZ July 50 selling for 1. Table 2-2 depicts the profitability of utilizing the July 40 or the July 50 for the covered writing. The in-the-money covered write of the July 40 affords 8 points, or nearly 18% protection down to price of 37 (the break-even point) at expiration. The out-of-the-money covered write of the July 50 offers only 1 point of downside protection at expiration. Hence, the *in-the-money covered write offers greater downside protection than does the out-of-the-money covered write.* This statement is true in general—not merely for this example.

In the balance of the financial world, it is normally true that investment positions offering less risk also have lower reward potential. The covered writing example just given is no exception. The in-the-money covered write of the July 40 has a maximum potential profit of $300 at any point above 40 at the time of expiration. However, the out-of-the-money covered write of the July 50 has a maximum potential profit of $600 at any point above 50 at expiration. *The maximum potential profit of an out-of-the-money covered write is generally greater than that of an in-the-money write.*

To make a true comparison between the two covered writes, one must look at what happens with the stock between 40 and 50 at expiration. The in-the-money write attains

TABLE 2-2.
Profit or loss of the July 40 and July 50 calls.

In-the-Money Write of July 40		Out-of-the-Money Write of July 50	
Stock at Expiration	Total Profit	Stock at Expiration	Total Profit
35	–$200	35	–$900
37	0	40	– 400
40	+ 300	44	0
45	+ 300	45	+ 100
50	+ 300	50	+ 600
60	+ 300	60	+ 600

its maximum profit anywhere within that range. Even a 5-point decline by the underlying stock at expiration would still leave the in-the-money writer with his maximum profit. However, *realizing the maximum profit potential with an out-of-the-money covered write always requires a rise in price by the underlying stock*. This further illustrates the more conservative nature of the in-the-money write. It should be noted that in-the-money writes, although having a smaller profit potential, can still be attractive on a percentage return basis, especially if the write is done in a margin account.

One can construct a more aggressive position by writing an out-of-the-money call. One's outlook for the underlying stock should be bullish in that case. If one is neutral or moderately bearish on the stock, an in-the-money covered write is more appropriate. If one is truly bearish on a stock he owns, he should sell the stock instead of establishing a covered write.

THE TOTAL RETURN CONCEPT OF COVERED WRITING

When one writes an out-of-the-money option, the overall position tends to reflect more of the result of the stock price movement and less of the benefits of writing the call. Since the premium on an out-of-the-money call is relatively small, the total position will be quite susceptible to loss if the stock declines. If the stock rises, the position will make money regardless of the result in the option at expiration. On the other hand, an in-the-money write is more of a "total" position—taking advantage of the benefit of the relatively large option premium. If the stock declines, the position can still make a profit; in fact, it can even make the maximum profit. Of course, an in-the-money write will also make money if the stock rises in price, but the profit is not generally as great in percentage terms as is that of an out-of-the-money write.

Those who believe in the *total return concept of covered writing* consider both downside protection and maximum potential return as important factors and are willing to have the stock called away, if necessary, to meet their objectives. When premiums are moderate or small, only in-the-money writes satisfy the total return philosophy.

Some covered writers prefer never to lose their stock through exercise, and as a result will often write options quite far out-of-the-money to minimize the chances of being called by expiration. These writers receive little downside protection and, to make money, must depend almost entirely on the results of the stock itself. Such a philosophy is more like being a stockholder and trading options against one's stock position than actually operating a covered writing strategy. In fact, some covered writers will attempt to buy back written options for quick profits if such profits materialize during the life of the covered write. This, too, is a stock ownership philosophy, not a covered writing strategy.

The total return concept represents the true strategy in covered writing, whereby one views the entire position as a single entity and is not predominantly concerned with the results of his stock ownership.

THE CONSERVATIVE COVERED WRITE

Covered writing is generally accepted to be a conservative strategy. This is because the covered writer always has less risk than a stockholder, provided that he holds the covered write until expiration of the written call. If the underlying stock declines, the covered writer will always offset part of his loss by the amount of the option premium received, no matter how small.

As was demonstrated in previous sections, however, some covered writes are clearly more conservative than others. Not all option writers agree on what is meant by a conservative covered write. Some believe that it involves writing an option (probably out-of-the-money) on a conservative stock, generally one with high yield and low volatility. It is true that the stock itself in such a position is conservative, but the position is more aptly termed a *covered write on a conservative stock*. This is distinctly different from a conservative covered write.

A true *conservative covered write* is one in which the total position is conservative—offering reduced risk and a good probability of making a profit. An in-the-money write, even on a stock that itself is not conservative, can become a conservative total position when the option itself is properly chosen. Clearly, an investor cannot write calls that are too deeply in-the-money. If he did, he would get large amounts of downside protection, but his returns would be severely limited. If all that one desired was maximum protection of his money at a nominal rate of profit, he could leave the money in a bank. Instead, the conservative covered writer strives to make a potentially acceptable return while still receiving an above-average amount of protection.

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Example: Again assume XYZ common stock is selling at 45 and an XYZ July 40 call is selling at 8. A covered write of the XYZ July 40 would require, in a cash account, an investment of \$3,700—\$4,500 to purchase 100 shares of XYZ, less the \$800 received in option premiums. The write has a maximum profit potential of \$300. The potential return from this position is therefore \$300/\$3,700, just over 8% for the period during which the write must be held. Since it is most likely that the option has 9 months of life or less, this return would be well in excess of 10% on a per annum basis. If the write were done in a margin account, the return would be considerably higher.

Note that we have ignored dividends paid by the underlying stock and commission charges, factors that are discussed in detail in the next section. Also, one should be aware that if he is looking at an *annualized return* from a covered write, there is no guarantee that such a return could actually be obtained. All that is certain is that the writer could make 8% in 9 months. There is no guarantee that 9 months from now, when the call expires, there will be an equivalent position to establish that will extend the same return for the remainder of the annualization period. Annual returns should be used only for comparative purposes between covered writes.

The writer has a position that has an annualized return (for comparative purposes) of over 10% and 8 points of downside protection. Thus, the *total position* is an investment that will not lose money unless XYZ common stock falls by more than 8 points, or about 18%; and is an investment that could return the equivalent of 10% annually should XYZ common stock rise, remain the same, or fall by 5 points (to 40). This is a conservative position. Even if XYZ itself is not a conservative stock, the action of writing this option has made the *total position* a conservative one. The only factor that might detract from the conservative nature of the total position would be if XYZ were so volatile that it could easily fall more than 8 points in 9 months.

In a strategic sense, the total position described above is better and more conservative than one in which a writer buys a conservative stock—yielding perhaps 6 or 7%—and writes an out-of-the-money call for a minimal premium. If this conservative stock were to fall in price, the writer would be in danger of being in a loss situation, because here the option is not providing anything more than the most minimal downside protection. As was described earlier, a high-yielding, low-volatility stock will not have much time premium in its in-the-money options, so that one cannot effectively establish an in-the-money write on such a "conservative" stock.

COMPUTING RETURN ON INVESTMENT

Now that the reader has some general feeling for covered call writing, it is time to discuss the specifics of computing return on investment. One should always know exactly what his potential returns are, including all costs, when he establishes a covered writing position. Commission rates are highly variable, depending on whether one uses a discount, electronic broker or not. In the examples that follow, it will be arbitrarily assumed that commissions are 3 cents per share for stock and $5 for an option contract. Individuals should use their own commission rates when actually calculating returns. Once the procedure for computing returns is clear, one can more logically decide which covered writes are the most attractive.

There are three basic elements of a covered write that should be computed before entering into the position. The first is the *return if exercised*. This is the return on investment that one would achieve if the stock were called away. For an out-of-the-money covered write, it is necessary for the stock to rise in price in order for the return if exercised to be achieved. However, for an in-the-money covered write, the return if exercised would be attained even if the stock were unchanged in price at option expiration. Thus, it is often advantageous to compute the *return if unchanged*—that is, the return that would be realized if the underlying stock were unchanged when the option expired. One can more fairly compare out-of-the-money and in-the-money covered writes by using the return if unchanged, since no assumption is made concerning stock price movement. The third important statistic that the covered writer should consider is the exact *downside break-even point* after all costs are included. Once this downside break-even point is known, one can readily compute the percentage of *downside protection* that he would receive from selling the call.

Example 1: An investor is considering the following covered write of a 6-month call: Buy 500 XYZ common at 43, sell 5 XYZ July 45 calls at 3. One must first compute the net investment required (Table 2-3). In a cash account, this investment consists of paying for the stock in full, less the net proceeds from the sale of the options. Note that this net investment figure includes all commissions necessary to establish the position. (The commissions used here are approximations, as they vary from firm to firm.) Of course, if the investor withdraws the option premium, as he is free to do, his net investment will consist of the stock cost plus commissions. Once the necessary investment is known, the writer can compute the return if exercised. Table 2-4 illustrates the computation. One first computes the profit if exercised and then divides that quantity by the net investment to obtain the return if exercised. Note that dividends are included in this computation; it is assumed that XYZ stock will pay $500 in dividends on the 500 shares during the life of the call. Moreover, all commissions are included as well—the net investment includes the original stock purchase and option sale commissions, and the stock sale commission is explicitly listed.

For the return computed here to be realized, XYZ stock would have to rise in price from its current price of 43 to any price above 45 by expiration. As noted earlier, it may be more useful to know what return could be made by the writer if the stock did not move anywhere at all. Table 2-5 illustrates the method of computing the *return if unchanged*—also called the *static return* and sometimes *incorrectly* referred to as the "expected return." Again, one first calculates the profit and then calculates the return by dividing the profit by the net investment. An important point should be made here: There is no stock sale commission included in Table 2-5. This is the most common way of calculating the return if unchanged; it is done this way because in a majority of cases, one would continue to hold the stock if it were unchanged and would write another call option against the same stock. Recall again, though, that *if the written call is in-the-money, the return if unchanged is the same as the return if exercised*. Stock sale commissions must therefore be included in that case.

TABLE 2-3.
Net investment required—cash account.

Stock cost (500 shares at 43)		$21,500
Plus stock purchase commissions	+	15
Less option premiums received	–	1,500
Plus option sale commissions	+	25
Net cash investment		$20,040

TABLE 2-4.
Return if exercised—cash account.

Stock sale proceeds (500 shares at 45)		$22,500
Less stock sale commissions	–	15
Plus dividends earned until expiration	+	500
Less net investment	–	20,040
Net profit if exercised		$ 2,945

$$\text{Return if exercised} = \frac{\$2,945}{\$20,040} = 14.7\%$$

TABLE 2-5.
Return if unchanged—cash account.

Unchanged stock value (500 shares at 43)		$21,500
Plus dividends	+	500
Less net investment	–	20,040
Profit if unchanged		$ 1,960

$$\text{Return if unchanged} = \frac{\$1,960}{\$20,040} = 9.8\%$$

Once the necessary returns have been computed and the writer has a feeling for how much money he could make in the covered write, he next computes the exact *downside break-even point* to determine what kind of *downside protection* the written call provides (Table 2-6). The total return concept of covered writing necessitates viewing both potential income *and* downside protection as important criteria for selecting a writing position. If the stock were held to expiration and the $500 in dividends received, the writer would break even at a price of 39.08. Again, a stock sale commission is not generally included in the break-even point computation, because the written call would expire totally worthless

TABLE 2-6.
Downside break-even point—cash account.

Net investment	$20,040
Less dividends	− 500
Total stock cost to expiration	$19,540
Divide by shares held	÷ 500
Break-even price	39.08

and the writer might then write another call on the same stock. Later, we discuss the subject of continuing to write against stocks already owned. It will be seen that in many cases, it is advantageous to continue to hold a stock and write against it again, rather than to sell it and establish a covered write in a new stock.

Next, we translate the break-even price into *percent downside protection* (Table 2-7), which is a convenient way of comparing the levels of downside protection among variously priced stocks. We will see later that it is actually better to compare the downside protection with the *volatility* of the underlying stock. It makes no sense to quote percent protection without knowing the volatility of the underlying stock. For example, 10% protection on AT&T is quite a bit more protection than the same percentage on a much more volatile stock, like Google, say, because Google is much more volatile than AT&T. The formula for computing protection in terms of volatility is in Chapter 28 (Mathematical Applications).

Before moving on to discuss what kinds of returns one should attempt to strive for in which situations, the same example will be worked through again for a covered write in a margin account. The use of margin will provide higher potential returns, since the net investment will be smaller. However, the margin interest charge incurred on the debit balance (the amount of money borrowed from the brokerage firm) will cause the break-even point to be higher, thus slightly reducing the amount of downside protection available from writing the call. Again, all commissions to establish the position are included in the net investment computation.

TABLE 2-7
Percent downside protection—cash account.

Initial stock price	43
Less break-even price	− 39.08
Points of protection	3.92
Divide by original stock price	÷ 43
Equals percent downside protection	9.1%

Example 2: Recall that the net investment for the cash write was \$20,040. A margin covered write requires less than half of the investment of a cash write when the margin rate (set by the Federal Reserve) is 50%. In a margin account, if one desires to remove the premium from the account, he may do so immediately provided that he has enough reserve equity in the account to cover the purchase of the stock. If he does so, his net investment would be equal to the debit balance calculation shown on the right in Table 2-8.

TABLE 2-8.
Net investment required—margin account.

Stock cost	\$21,500		
Plus stock commissions	+ 15	Debit balance calculation:	
Net stock cost	\$21,515	Net stock cost	\$21,515
Times margin rate	× 50%	Less equity	– 10,758
Equity required	\$10,758	Debit balance (at 50% margin)	\$10,757
Less premiums received	– 1,500		
Plus option commissions	+ 25		
Net margin investment	\$ 9,283		

Tables 2-9 to 2-12 illustrate the computation of returns from writing on margin. If one has already computed the cash returns, he can use method 2 most easily. Method 1 involves no prior profit calculations.

TABLE 2-9.
Return if exercised—margin account.

Method 1		Method 2	
Stock sale proceeds	\$22,500	Net profit if exercised—cash	\$2,945
Less stock commission	– 15	Less margin interest charges	– 538
Plus dividends	+ 500	Net profit if exercised—margin	\$2,407
Less margin interest charges (10% on \$10,758 for 6 months)	– 538		
Less debit balance	– 10,757		
Less net margin investment	– 9,283		
Net profit—margin	\$ 2,407		

$$\text{Return if exercised} = \frac{\$2,407}{\$9,283} = 25.9\%$$

TABLE 2-10.
Return if unchanged—margin account.

Method 1		Method 2	
Unchanged stock value (500 shares at 43)	$21,500	Profit if unchanged—cash	$1,960
Plus dividends	+ 500	Less margin interest charges	– 538
Less margin interest charges (10% on $10,910 debit for 6 months)	– 538	Net profit if unchanged—margin	$1,422
Less debit balance	– 10,757		
Less net investment (margin)	– 9,283		
Net profit if unchanged—margin	$ 1,422		

$$\text{Return if unchanged} = \frac{\$1{,}422}{\$9{,}283} = 15.3\%$$

TABLE 2-11.
Break-even point—margin write.

Net margin investment	$ 9,283
Plus debit balance	+ 10,757
Less dividends	– 500
Plus margin interest charges	+ 538
Total stock cost to expiration	$20,078
Divide by shares held	÷ 500
Break-even point—margin	40.16

The return if exercised is 25.6% for the covered write using margin. In Example 1 the return if exercised for a cash write was computed as 14.7%. Thus, the return if exercised from a margin write is considerably higher. In fact, unless a fairly deep in-the-money write is being considered, the return on margin will always be higher than the return from cash. The farther out-of-the-money that the written call is, the bigger the discrepancy between cash and margin returns will be when the return if exercised is computed.

TABLE 2-12.
Percent downside protection—margin write.1

Initial stock price	43
Less break-even price—margin	– 40.16
Points of protection	2.84
Divide by original stock price	÷ 43
Equals percent downside protection—margin	6.6%

As with the computation for return if exercised for a write on margin, the return if unchanged calculation is similar for cash and margin also. The only difference is the subtraction of the margin interest charges from the profit. The return if unchanged is also higher for a margin write, provided that there is enough option premium to compensate for the margin interest charges. The return if unchanged in the cash example was 9.8% versus 15.3% for the margin write. In general, the father from the strike in either direction—out-of-the-money or in-the-money—the less the return if unchanged on margin will exceed the cash return if unchanged. In fact, for deeply out-of-the-money or deeply in-the-money calls, the return if unchanged will be higher on cash than on margin. Table 2-11 shows that the break-even point on margin, 40.16, is higher than the break-even point from a cash write, 39.08, because of the margin interest charges. Again, the percent downside protection can be computed as shown in Table 2-12. Obviously, since the break-even point on margin is higher than that on cash, there is less percent downside protection in a margin covered write.

One other point should be made regarding a covered write on margin: The brokerage firm will loan you only *half of the strike price amount* as a maximum. Thus, it is *not* possible, for example, to buy a stock at 20, sell a deeply in-the-money call struck at 10 points, and trade for free. In that case, the brokerage firm would loan you only 5—half the amount of the strike.

Even so, it is still possible to create a covered call write on margin that has little or even *zero* margin requirement. For example, suppose a stock is selling at 38 and that a long-term LEAPS option struck at 40 is selling for 19. Then the margin requirement is zero! This does not mean you're getting something for free, however. True, your investment is zero, but your *risk* is still 19 points. Also, your broker would ask for some sort of minimum margin to begin with and would of course ask for maintenance margin if the underlying stock should fall in price. Moreover, you would be paying margin interest all during the life of this long-term LEAPS option position. Leverage can be a good thing or a bad thing, and this strategy has a great deal of leverage. So be careful if you utilize it.

COMPOUND INTEREST

The astute reader will have noticed that our computations of margin interest have been overly simplistic; the compounding effect of interest rates has been ignored. That is, since interest charges are normally applied to an account monthly, the investor will be paying interest in the later stages of a covered writing position not only on the original debit, but on all previous monthly interest charges. This effect is described in detail in a later chapter on arbitrage techniques. Briefly stated, rather than computing the interest charge as the debit times the interest rate multiplied by the time to expiration, one should technically use:

$$\text{Margin interest charges} = \text{Debit}\ [(1 + r)^t - 1]$$

where r is the interest rate per month and t the number of months to expiration. (It would be incorrect to use days to expiration, since brokerage firms compute interest monthly, not daily.)

In Example 2 of the preceding section, the debit was $10,757, the time was 6 months, and the annual interest rate was 10%. Using this more complex formula, the margin interest charges would be $549, as opposed to the $538 charge computed with the simpler formula. Thus, the difference is usually small, in terms of percentage, and *it is therefore common practice to use the simpler method.*

SIZE OF THE POSITION

If one is trading at a firm that does not have a minimum fixed commission cost (i.e., where the per-share cost is the same, no matter how few shares are bought), then the number of shares in the covered write is irrelevant to the returns. However, most brokers charge a minimum commission, even the deep-discount electronic ones. Furthermore a full service broker often charges a lower per-share commission rate for larger trades (and, by inference, a higher per-share commission rate for smaller trades).

The covered writer should keep this in mind, for a lower per-share commission cost results in higher returns, of course. Even traders using deep discount brokers should make certain that they are establishing the covered write with enough shares so that the broker's minimum commission charge is exceeded. Buying too few shares for covered writing purposes can lower returns considerably, if the minimum commission charge comes into play.

WHAT A DIFFERENCE A DIME MAKES

Another aspect of covered writing that can be important as far as potential returns are concerned is, of course, the prices of the stock and option involved in the write. It may seem insignificant that one has to pay an extra few cents for the stock or possibly receives a dime or 20 cents less for the call, but even a relatively small fraction can alter the potential returns by a surprising amount. This is especially true for in-the-money writes, although any write will be affected. Let us use the previous 500-share covered writing example, again including all costs.

As before, the results are more dramatic for the margin write than for the cash write. In neither case does the break-even point change by much. However, the potential returns are altered significantly. Notice that if one pays an extra dime for the stock and receives a dime less for the call—the far right-hand column in Table 2.13—he may greatly negate the effect of writing against a large number of shares. From Table 2.13, one can see that writting against 300 shares at those prices (43 for the stock and 3 for the call) is approximately the same return as writing against *500 shares* if the stock costs 43.10 and the option brings in 2.90.

Table 2.13 should clearly demonstrate that entering a covered writing order *at the market* may not be a prudent thing to do, especially if one's calculations for the potential returns are based on last sales or on closing prices in the newspaper. In the next section, we discuss in depth the proper procedure for entering a covered writing order.

TABLE 2.13.
Effect of stock and option prices on writing returns.

	Buy Stock at 43 Sell Call at 3	Buy Stock at 43.10 Sell Call at 3	Buy Stock at 43.10 Sell Call at 2.90
Return if exercised	14.7% cash	14.4% cash	14.1% cash
	25.9% margin	25.3% margin	24.6% margin
Return if unchanged	9.8% cash	9.8% cash	9.5% cash
	15.3% margin	15.3% margin	14.7% margin
Break-even point	39.08 cash	39.18 cash	39.28 cash
	40.16 margin	40.26 margin	40.36 margin

EXECUTION OF THE COVERED WRITE ORDER

When establishing a covered writing position, the question often arises: Which should be done first—buy the stock or sell the option? The correct answer is that neither should be done first! In fact, *a simultaneous transaction of buying the stock and selling the option is the only way of assuring that both sides of the covered write are established at desired price levels.*

If one "legs" into the position—that is, buys the stock first and then attempts to sell the option, or vice versa—he is subjecting himself to a risk.

Example: An investor wants to buy XYZ at 43 and sell the July 45 call at 3. If he first sells the option at 3 and then tries to buy the stock, he may find that he has to pay more than 43 for the stock. On the other hand, if he tries to buy the stock first and *then* sell the option, he may find that the option price has moved down. In either case the writer will be accepting a lower return on his covered write. Table 2.13 demonstrated how one's returns might be affected if he has to give up a dime by "legging" into the position.

ESTABLISHING A NET POSITION

What the covered writer really wants to do is ensure that his net price is obtained. If he wants to buy stock at 43 and sell an option at 3, he is attempting to establish the position at 40 *net*. He normally would not mind paying 43.10 for the stock if he can sell the call at 3.10, thereby still obtaining 40 net.

A *"net" covered writing order* must be placed either directly with a brokerage firm's option desk or through the "spread order entry," if dealing with an online broker. If you are planning on doing a lot of covered writing, make sure your brokerage firm offers the placement of "net" orders—either online or directly through an order desk. This is also referred to as a contingent order. Most major brokerage firms offer this service to their clients, although some place a minimum number of shares on the order. That is, one must write against at least 500 or 1,000 shares in order to avail himself of the service. There are, however, brokerage firms that will take net orders even for 100-share covered writes. Since the chances of giving away a dime are relatively great if one attempts to execute his own order by placing separate orders on two exchanges—stock and option—he should avail himself of the broker's service. Moreover, if his orders are for a small number of shares, he should deal with a broker who will take net orders for small positions.

The reader must understand that *there is no guarantee that a net order will be filled.* The net order is always a "not held" order, meaning that the customer is not guaranteed an execution even if it appears that the order could be filled at prevailing market bids and offers. Of course, the broker will attempt to fill the order if it can reasonably be accomplished, since that is his livelihood.

If one buys stock at 43 and sells the call at 3, is the return really the same as buying the stock at 43.10 and selling the call at 3.10? The answer is, yes, the returns are very similar when the prices differ by small amounts. This can be seen without the use of a table. If one pays a dime more for the stock, his investment increases by $10 per 100 shares, or $50 total on a 500-share transaction. However, the fact that he has received an extra dime for the call means that the investment is reduced by $50. Thus, there is no effect on the net investment except for commissions. The commission on 500 shares at 43.10 may be slightly higher than the commission for 500 shares at 43. Similarly, the commission on 5 calls at 3.10 may be slightly higher than that on 5 calls at 3. Even so, the increase in commissions would be so small that it would not affect the return by more than one-tenth of 1%.

To carry this concept to extremes may prove somewhat misleading. If one were to buy stock at 40.50 and sell the call at .50, he would still be receiving 40 net, but several aspects would have changed considerably. The return if exercised remains amazingly constant, but the return if unchanged and the percentage downside protection are reduced dramatically. If one were to buy stock at 48 and sell the call at 8—again for 40 net—he would improve the return if unchanged and the percentage downside protection. In reality, when one places a "net" order with a brokerage firm, he normally gets an execution with prices quite close to the ones at the time the order was first entered. It would be a rare case, indeed, when either upside or downside extremes such as those mentioned here would occur in the same trading day.

SELECTING A COVERED WRITING POSITION

The preceding sections, in describing types of covered writes and how to compute returns and break-even points, have laid the groundwork for the ultimate decision that every covered writer must make: choosing which stock to buy and which option to write. This is not necessarily an easy task, because there are large numbers of stocks, striking prices, and expiration dates to choose from.

Since the primary objective of covered writing for most investors is increased income through stock ownership, the return on investment is an important consideration in determining which write to choose. However, the decision must not be made on the basis of return alone. More volatile stocks will offer higher returns, but they may also involve more risk because of their ability to fall in price quickly. Thus, the amount of downside protection is the other important objective of covered writing. Finally, the quality and technical or fundamental outlook of the underlying stock itself are of importance as well. The following section will help to quantify how these factors should be viewed by the covered writer.

PROJECTED RETURNS

The return that one strives for is somewhat a matter of personal preference. In general, *the annualized return if unchanged should be used as the comparative measure between various covered writes.* In using this return as the measuring criterion, one does not make any assumptions about the stock moving up in price in order to attain the potential return. A general rule used in deciding what is a minimally acceptable return is to consider a covered writing position only when the return if unchanged is at least 1% per month. That is, a 3-month write would have to offer a return of at least 3% and a 6-month write would have to have a return if unchanged of at least 6%. During periods of expanded option premiums, there may be so many writes that satisfy this criterion that one would want to raise his sights somewhat, say to 1½% or 2% per month. Also, one must feel personally comfortable that his minimum return criterion—whether it be 1% per month or 2% per month—is large enough to compensate for the risks he is taking. That is, the downside risk of owning stock, should it fall far enough to outdistance the premium received, should be adequately compensated for by the potential return. It should be pointed out that 1% per month is not a return to be taken lightly, especially if there is a reasonable assurance that it can be attained. However, if less risky investments, such as bonds, were yielding 12% annually, the covered writer must set his sights higher.

Normally, the returns from various covered writing situations are compared by annualizing the returns. One should not, however, be deluded into believing that he can always attain the projected annual return. A 6-month write that offers a 6% return annualizes to 12%. But if one establishes such a position, all that he can achieve is 6% in 6 months. One does not really know for sure that 6 months from now there will be another position available that will provide 6% over the next 6 months.

The deeper that the written option is in-the-money, the higher the probability that the return if unchanged will actually be attained. In an in-the-money situation, recall that the return if unchanged is the same as the return if exercised. Both would be attained unless the stock fell below the striking price by expiration. Thus, for an in-the-money write, the projected return is attained if the stock rises, remains unchanged, or even falls slightly by the time the option expires. Higher potential returns are available for out-of-the-money writes if the stock rises. However, should the stock remain the same or decline in price, the out-of-the-money write will generally underperform the in-the-money write. This is why the return if unchanged is a good comparison.

DOWNSIDE PROTECTION

Downside protection is more difficult to quantify than projected returns are. As mentioned earlier, the percentage of downside protection is often used as a measure. This is somewhat misleading, however, since the more volatile stocks will always offer a large percentage of downside protection (their premiums are higher). The difficulty arises in trying to decide if 10% protection on a volatile stock is better than or worse than, say, 6% protection on a less volatile stock. There are mathematical ways to quantify this, but because of the relatively advanced nature of the computations involved, they are not discussed until later in the text, in Chapter 28 on mathematical applications.

Rather than go into involved mathematical calculations, many covered writers use the percentage of downside protection and will only consider writes that offer a certain minimum level of protection, say 10%. Although this is not exact, it does strive to ensure that one has minimal downside protection in a covered write, as well as an acceptable return. A standard figure that is often used is the 10% level of protection. Alternatively, one may also require that the write be a certain percent in-the-money, say 5%. This is just another way of arriving at the same concept.

THE IMPORTANCE OF STRATEGY

In a conservative option writing strategy, one should be looking for minimum returns if unchanged of 1% per month, with downside protection of at least 10%, as general guidelines. Employing such criteria automatically forces one to write in-the-money options in line with the total return concept. The overall position constructed by using such guidelines as these will be a relatively conservative position—regardless of the volatility of the underlying stock—since the levels of protection will be large but a reasonable return can still be attained. There is a danger, however, in using fixed guidelines, because market conditions change. In the early days of listed options, premiums were so large that virtually every at- or in-the-money covered write satisfied the foregoing criteria. However, now one should work with a ranked list of covered writing positions, or perhaps two lists. A daily computer ranking of either or both of the following categories would help establish the most attractive types of conservative covered writes. One list would rank, by annualized return, the writes that afford, as a minimum, the desired downside protection level, say 10%. The other list would rank, by percentage downside protection, all the writes that meet at least the minimum acceptable return if unchanged, say 12%. If premium levels shrink and the lists become quite small on a daily basis, one might consider expanding the criteria to view more potential situations. On the other hand, if premiums expand dramatically, one might consider using more restrictive criteria, to reduce the number of potential writing candidates.

A different group of covered writers may favor a more aggressive strategy of out-of-the-money writes. *There is some mathematical basis to believe, in the long run, that moderately out-of-the-money covered writes will perform better than in-the-money writes.* In falling or static markets, any covered writer, even the more aggressive one, will outperform the stockowner who does not write calls. The out-of-the-money covered writer has more risk in such a market than the in-the-money writer does. But in a rising market, the out-of-the-money covered writer will not limit his returns as much as the in-the-money writer will. As stated earlier, the out-of-the-money writer's performance will more closely follow the performance of the underlying stock; that is, it will be more volatile on a quarter-by-quarter basis.

There is merit in either philosophy. The in-the-money writes appeal to those investors looking to earn a relatively consistent, moderate rate of return. This is the *total return concept*. These investors are generally concerned with preservation of capital, thus striving for the greater levels of downside protection available from in-the-money writes. On the other hand, some investors prefer to strive for higher potential returns through writing out-of-the-money calls. These more aggressive investors are willing to accept more downside risk in their covered writing positions in exchange for the possibility of higher returns should the underlying stock rise in price. These investors often rely on a bullish research opinion on a stock in order to select out-of-the-money writes.

Although the type of covered writing strategy pursued is a matter of personal philosophy, it would seem that the benefits of in-the-money strategy—more consistent returns and lessened risk than stock ownership will normally provide—would lead the portfolio manager or less aggressive investor toward this strategy. If the investor is interested in achieving higher returns, some of the strategies to be presented later in the book may be able to provide higher returns with less risk than can out-of-the-money covered writing.

The final important consideration in selecting a covered write is the underlying stock itself. One does not necessarily have to be bullish on the underlying stock to take a covered writing position. As long as one does not foresee a potential decline in the underlying stock, he can feel free to establish the covered writing position. It is generally best if one is neutral or slightly bullish on the underlying stock. *If one is bearish, he should not take a covered writing position on that stock*, regardless of the levels of protection that can be obtained. An even broader statement is that one should not establish a covered write on a stock that he does not want to own. Some individual investors may have qualms about buying stock they feel is too volatile for them. Impartially, if the return and protection are adequate, the characteristics of the total position are different from those of the underlying stock. However, it is still true that one should not invest in positions that he considers too risky for his portfolio, nor should one establish a covered write just because he likes a particular stock. If the potential return is unchanged or levels of downside protection do not meet one's criteria, the write should not be established.

The covered writing *strategist* strives for a balance between acceptable returns and downside protection. He rejects situations that do not meet his criteria in either category and rejects stocks on which he is bearish. The resulting situations will probably fulfill the objectives of a conservative covered writing program: increased income, protection, and less variability of results on a less volatile investment portfolio.

WRITING AGAINST STOCK ALREADY OWNED

Establishing covered writing positions against stock that has previously been purchased involves other factors. It is often the case that an investor owns stock that has listed options trading, but feels that the returns from writing are too low in comparison to other covered writes that simultaneously exist in the marketplace. This opinion may be valid, but often arises from the fact that the investor has seen a computer-generated list showing returns on his stock as being low in comparison to similarly priced stocks. One should note that such lists generally assume that stock is bought in order to establish the covered write: the returns are usually not computed and published for writing against stock already held. It may be the case that the commission costs for selling one stock and investing in another may alter the returns so substantially that one would be better off to write against the shares of stock initially held.

Example: An investor owns XYZ stock and is comparing it against AAA stock for writing purposes. If AAA is more volatile than XYZ, the current prices might appear as follows:

Stock	Oct 50 Call
XYZ: 50	4
AAA: 50	6

Without going into as much detail, the other significant aspects of these two writes are:

	XYZ	AAA
Return if exercised—margin	9.7%	13.4%
Downside break-even point—cash	45.58	44.08
Downside break-even point—margin	46.83	45.33

Seeing these calculations, the XYZ stockholder may feel that it is not advisable to write against his stock, or he may even be tempted to sell XYZ and buy AAA in order to establish a covered write. Either of these actions could be a mistake.

When commission costs were higher, in the early years of listed option trading, one would be reluctant to switch out of a stock that he already owned to achieve larger returns in a stock that he didn't own, for there would be rather onerous stock commissions on both the buy and the sell. In fact, if the reader is paying a rather large commission rate, then he should carefully assess the returns from the two writes. He should calculate returns on the XYZ stock as if no commission were paid upon entry and compare *those* with the returns on AAA stock. For most investors, though, commissions are rather low today, so there isn't much of an impediment to switch from XYZ to AAA in that regard.

However, commissions are not the only consideration. Each investor must decide for himself whether it is worth the increase in return to switch to AAA, because he would be switching to a more volatile stock that doesn't pay a dividend. Yes, the covered writing static returns might justify it, but a volatility-based calculation of the risk would be necessary to make the final decision.

This same logic might apply in situations where an investor has been doing covered writing. If he owns stock on which an option has expired, he will have to decide whether to write against the same stock again or to sell the stock and buy a new stock. In the high-commission days, he might have decided to write against the same stock, even if the returns weren't nearly as attractive going forward, merely because of the large commission costs of selling one stock and buying another. But in the modern era of low commission rates, this really shouldn't be a concern. The writer should pursue the best overall total return covered write. In fact, it can be a lethargic mistake to get lured into just writing against the same stock, month after month, or quarter after quarter, even if the returns have diminished.

A WORD OF CAUTION

The stockholder who owns stock from a previous purchase and later contemplates writing calls against that stock must be aware of his situation. He must realize and accept the fact that he might lose his stock via assignment. If he is determined to retain ownership of the stock, he may have to buy back the written option at a loss should the underlying stock increase in price. In essence, he is limiting the stock's upside potential. If a stockholder is going to be frustrated and disappointed when he is not fully participating during a rally in his stock, he should not write a call in the first place. Perhaps he could utilize the incremental return concept of covered writing, a topic covered later in this chapter.

As stressed earlier, a covered writing strategy involves viewing the stock and option as a *total* position. *It is not a strategy wherein the investor is a stockholder who also trades options against his stock position.* If the stockholder is selling the calls because he thinks the stock is going to decline in price and the call trade itself will be profitable, he may be putting himself in a tenuous position. Thinking this way, he will probably be satisfied only if he makes a profit on the call trade, regardless of the unrealized result in the underlying stock. This sort of philosophy is contrary to a covered writing strategy philosophy. Such an investor—he is really becoming a trader—should carefully review his motives for writing the call and anticipate his reaction if the stock rises substantially in price after the call has been written.

In essence, *writing calls against stock that you have no intention of selling is tantamount to writing naked calls!* If one is going to be extremely frustrated, perhaps even experiencing sleepless nights, if his stock rises above the strike price of the call that he has written, then he is experiencing trials and tribulations much as the writer of a naked call would if the same stock move occurred. This is an unacceptable level of emotional worry for a true covered writing strategist.

Think about it. If you have some very low-cost-basis stock that you don't really want to sell, and then you sell covered calls against that stock, what do you wish will happen? Most certainly you wish that the options will expire worthless (i.e., that the stock won't get called away)—exactly what a naked writer wishes for.

The problems can be compounded if the stock rises, and one then decides to roll these calls. Rather than spend a small debit to close out a losing position, an investor may attempt to roll to more distant expiration months and higher strike prices in order to keep bringing in credits. Eventually, he runs out of room as the lower strikes disappear, and he has to either sell some stock or pay a *big* debit to buy back the written calls. So, if the underlying stock continues to run higher, the writer suffers emotional devastation as he attempts to "fight the market." There have been some classic cases of Murphy's law whereby people have covered the calls at a big debit rather than let their "untouchable" stock be called away, just before the stock itself or the stock market collapsed.

One should be very cautious about writing covered calls against stocks that he doesn't intend to sell. If one feels that he cannot sell his stock, for whatever reason—tax considerations, emotional ties, etc.—he really should *not* sell covered calls against it. Perhaps buying a protective put (discussed in a later chapter) would be a better strategy for such a stockholder.

DIVERSIFYING RETURN AND PROTECTION IN A COVERED WRITE

FUNDAMENTAL DIVERSIFICATION TECHNIQUES

Quite clearly, the covered writing strategist would like to have as much of a combination of high potential returns and adequate downside protection as he can obtain. Writing an out-of-the-money call will offer higher returns if exercised, but it usually affords only a modest amount of downside protection. On the other hand, writing an in-the-money call will provide more downside cushion but offers a lower return if exercised. For some strategists, this diversification is realized in practice by writing out-of-the-money calls on some stocks and in-the-moneys on other stocks. There is no guarantee that writing in this manner on a list of diversified stocks will produce superior results. One is still forced to pick the stocks that he expects will perform better (for out-of-the-money writing), and that is difficult to do. Moreover, the individual investor may not have enough funds available to diversify into many such situations. There is, however, another alternative to obtaining diversification of both returns and downside protection in a covered writing situation.

The writer may often do best by writing half of his position against in-the-moneys and half against out-of-the-moneys on the same stock. This is especially attractive for a stock whose out-of-the-money calls do not appear to provide enough downside protection, and at the same time, whose in-the-money calls do not provide quite enough return. By writing both options, the writer may be able to acquire the return and protection diversification that he is seeking.

Example: The following prices exist for 6-month calls:

XYZ common stock, 42;
XYZ April 40 call, 4; and
XYZ April 45 call, 2.

The writer wishing to establish a covered write against XYZ common stock may like the protection afforded by the April 40 call, but may not find the return particularly attractive. He may be able to improve his return by writing April 45's against part of his position. Assume the writer is considering buying 1,000 shares of XYZ. Table 2.14 compares the attributes of writing the out-of-the-money (April 45) only, or of writing only the in-the-money (April 40), or of writing 5 of each. The table is based on a cash covered write, but returns and protection would be similar for a margin write. Commissions are included in the figures.

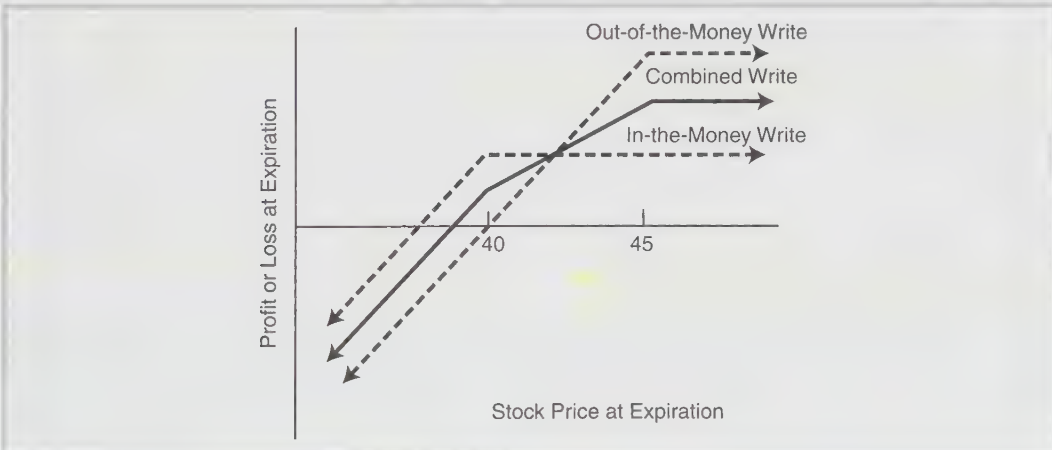
It is easily seen that the *"combined" write*—half of the position against the April 40's and the other half against the April 45's—offers the best balance of return and protection. The in-the-money call, by itself, provides over 11% downside protection, but the 7.6% return if exercised is just a bit more than 1% per month. Thus, one might not want to write April 40's against his entire position, because the potential return is small. At the same time, the April 45's, if written against the entire stock position, would provide for an attractive return if exercised (over 2% per month) but offer only 7% downside protection. The combined write, which has the better features of both options, offers over 11% return if exercised (nearly 1.5% per month) and affords over 8% downside protection. By writing both calls, the writer has potentially solved the problems inherent in writing entirely out-of-the-moneys or entirely in-the-moneys. The "combined" write frees the covered writer from having to initially take a bearish (in-the-money write) or bullish (out-of-the-money write) posture on the stock if he does not want to. This is often necessary on a low-volatility stock trading between striking prices.

For those who prefer a graphic representation, the profit graph shown in Figure 2-2 compares the combined write of both calls with either the in-the-money write or the out-of-the-money write (dashed lines). It can be observed that all three choices are equal if XYZ is near 42 at expiration; all three lines intersect there.

TABLE 2.14
Attributes of various writes.

	In-the-Money Write	Out-of-the-Money Write	Write Both Calls
Buy 1,000 XYZ and sell	10 April 40's	10 April 45's	5 April 40's and 5 April 45's
Return if exercised	7.6%	14.7%	11.2%
Return if unchanged	7.6%	7.3%	7.4%
Percent protection	11.7%	7.0%	9.3%

FIGURE 2-2.
Comparison: combined write vs. in-the-money write and out-of-the-money write.



This technique can be useful in providing diversification between protection and return, not only for an individual position but for a large part of a portfolio. In the modern era, with commissions on a per-share basis, the pertinent returns can be computed essentially by averaging the returns on the individual writes. For example, the return if exercised for the April 40s was 7.6%; the return if exercised for the April 45s is 14.7%. So the return for a write that utilizes half of each is the average, 11.2%.

OTHER DIVERSIFICATION TECHNIQUES

Holders of large positions in a particular stock may want even more diversification than can be provided by writing against two different striking prices. Institutions, pension funds, and large individual stockholders may fall into this category. It is often advisable for such large stockholders to *diversify* their writing *over time* as well as over at least two striking prices. By diversifying over time—for example, writing one-third of the position against near-term calls, one-third against middle-term calls, and the remaining third against long-term calls—one can gain several benefits. First, all of one's positions need not be adjusted at the same time. This includes either having the stock called away or buying back one written call and selling another. Moreover, one is not subject only to the level of option premiums that exist at the time one series of calls expires. For example, if one writes only 9-month calls and then rolls them over when they expire, he may unnecessarily be subjecting himself to the potential of lower returns. If option premium levels happen to be low when it is time for this 9-month call writer to sell more calls, he will be establishing a less-than-optimum write for up to 9 months. By spreading his writing out over time, he would, at worst, be subjecting only one-third of his holding to the low-premium write. Hopefully, premiums would expand before the next expiration 3 months later, and he would then be getting a relatively better premium on the next third of his portfolio. There is an important aside here: The individual or relatively small investor who owns only enough stock to write one series of options should generally not write the longest-term calls for this very reason. He may not be obtaining a particularly attractive level of premiums, but may feel he is forced to retain the position until expiration. Thus, he could be in a relatively poor write for as long as 9 months. Finally, this type of diversification may also lead to having calls at various striking prices as the market fluctuates cyclically. All of one's stock is not necessarily committed at one price if this diversification technique is employed.

This concludes the discussion of how to establish a covered writing position against stock. Covered writes against other types of securities are described later.

FOLLOW-UP ACTION

Establishing a covered write, or any option position for that matter, is only part of the strategist's job. Once the position has been taken, it must be monitored closely so that adjustments may be made should the stock drop too far in price. Moreover, even if the stock remains relatively unchanged, adjustments will need to be made as the written call approaches expiration.

Some writers take no follow-up action at all, preferring to let a stock be called away if it rises above the striking price at the expiration of the option, or preferring to let the original expire worthless if the stock is below the strike. These are not always optimum actions; there may be much more decision making involved.

Follow-up action can be divided into three general categories:

1. protective action to take if the stock drops,
2. aggressive action to take when the stock rises, or
3. action to avoid assignment if the time premium disappears from an in-the-money call.

There may be times when one decides to close the entire position before expiration or to let the stock be called away. These cases are discussed as well.

PROTECTIVE ACTION IF THE UNDERLYING STOCK DECLINES IN PRICE

The covered writer who does not take protective action in the face of a relatively substantial drop in price by the underlying stock may be risking the possibility of large losses. Since covered writing is a strategy with limited profit potential, one should also take care to limit losses. Otherwise, one losing position can negate several winning positions. The simplest form of follow-up action in a decline is to merely close out the position. This might be done if the stock declines by a certain percentage, or if the stock falls below a technical support level. Unfortunately, this method of defensive action may prove to be an inferior one. The investor will often do better to continue to sell more time value in the form of additional option premiums.

Follow-up action is generally taken by buying back the call that was originally written and then writing another call, with a different striking price and/or expiration date, in its place. Any adjustment of this sort is referred to as a *rolling action*. When the underlying stock drops in price, one generally buys back the original call—presumably at a profit since the underlying stock has declined—and then sells a call with a lower striking price. This is known as *rolling down*, since the new option has a lower striking price.

Example: The covered writing position described as "buy XYZ at 51, sell the XYZ January 50 call at 6" would have a maximum profit potential at expiration of 5 points. Downside protection is 6 points down to a stock price of 45 at expiration. These figures do not include commissions, but for the purposes of an elementary example, the commissions will be ignored.

If the stock begins to decline in price, taking perhaps two months to fall to 45, the following option prices might exist:

XYZ common, 45;
XYZ January 50 call, 1; and
XYZ January 45 call, 4.

The covered writer of the January 50 would, at this time, have a small unrealized loss of one point in his overall position: His loss on the common stock is 6 points, but he has a 5-point gain in the January 50 call. (This demonstrates that *prior to expiration*, a loss occurs at the "break-even" point.) If the stock should continue to fall from these levels, he could have a larger loss at expiration. The call, selling for one point, only affords one more point of downside protection. If a further stock price drop is anticipated, *additional downside protection can be obtained by rolling down*. In this example, if one were to buy back the January 50 call at 1 and sell the January 45 at 4, he would be rolling down. This would increase his protection by another three points—the credit generated by buying the 50 call at 1 and selling the 45 call at 4. Hence, his downside break-even point would be 42 after rolling down.

Moreover, if the stock were to remain unchanged—that is, if XYZ were exactly 45 at January expiration—the writer would make an *additional* $300. If he had not rolled down, the most *additional* income that he could make, if XYZ remained unchanged, would be the remaining $100 from the January 50 call. So *rolling down gives more downside protection against a further drop in stock price and may also produce additional income if the stock price stabilizes.*

In order to more exactly evaluate the overall effect that was obtained by rolling down in this example, one can either compute a profit table (Table 2-15) or draw a net profit graph (Figure 2-3) that compares the original covered write with the rolled-down position.

Note that the rolled-down position has a smaller maximum profit potential than the original position did. This is because, by rolling down to a January 45 call, the writer limits his profits anywhere above 45 at expiration. He has committed himself to sell stock 5 points lower than the original position, which utilized a January 50 call and thus had limited profits above 50. *Rolling down generally reduces the maximum profit potential of the covered write.* Limiting the maximum profit may be a secondary consideration, however, when a stock is breaking downward. Additional downside protection is often a more pressing criterion in that case.

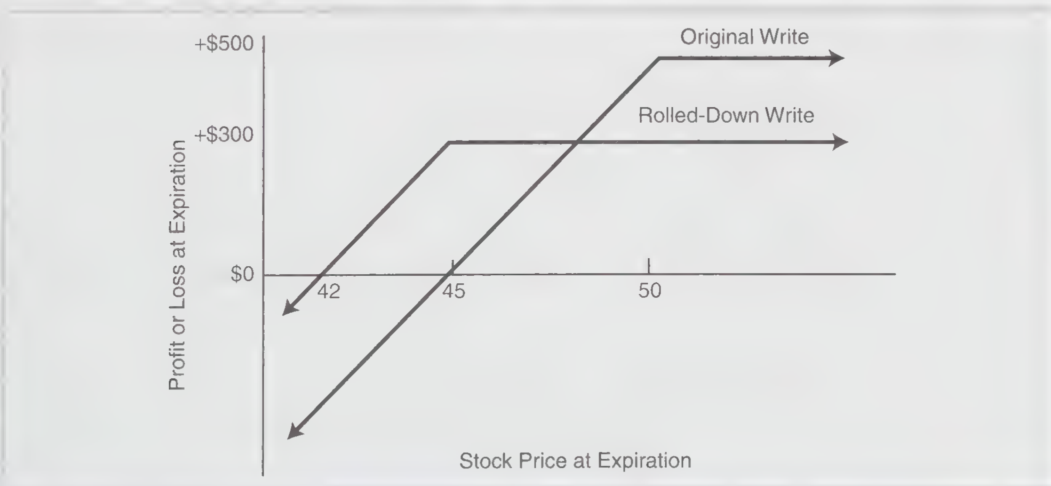
Anywhere below 45 at expiration, the rolled-down position does $300 better than the original position, because of the $300 credit generated from rolling down. In fact, the rolled-down position will outperform the original position even if the stock rallies back to, but not above, a price of 48. At 48 at expiration, the two positions are equal, both producing a $300 profit. If the stock should reverse direction and rally back above 48 by expiration, the writer would have been better off not to have rolled down. All these facts are clear from Table 2.15 and Figure 2-3.

Consequently, the only case in which it does not pay to roll down is the one in which the stock experiences a reversal—a rise in price after the initial drop. The selection of where to roll down is important, because rolling down too early or at an inappropriate

TABLE 2.15.
Profit table.

XYZ Price at Expiration	Profit from January 50 Write	Profit from Rolled Position
40	– $500	– 200
42	– 300	0
45	0	+ 300
48	+ 300	+ 300
50	+ 500	+ 300
60	+ 500	+ 300

FIGURE 2-3.
Comparison: original covered write vs. rolled-down write.



price could limit the returns. Technical support levels of the stock are often useful in selecting prices at which to roll down. If one rolls down after technical support has been broken, the chances of being caught in a stock-price-reversal situation would normally be reduced.

The above example is rather simplistic; in actual practice, more complicated situations may arise, such as a sudden and fairly steep decline in price by the underlying stock. This may present the writer with what is called a *locked-in loss*. This means, simply, that there is no option to which the writer can roll down that will provide him with enough premium to realize any profit if the stock were then called away at expiration. These situations arise more commonly on lower-priced stocks, where the striking prices are relatively far apart in percentage terms. Out-of-the-money writes are more susceptible to this problem than are in-the-money writes. Although it is not emotionally satisfying to be in an investment position that cannot produce a profit—at least for a limited period of time—it may still be beneficial to roll down to protect as much of the stock price decline as possible.

Example: For the covered write described as "buy XYZ at 20, sell the January 20 call at 2," the stock unexpectedly drops very quickly to 16, and the following prices exist:

XYZ common, 16;
XYZ January 20 call, .50; and
XYZ January 15 call, 2.50.

The covered writer is faced with a difficult choice. He currently has an unrealized loss of 2.50 points—a 4-point loss on the stock which is partially offset by a 1.50-point gain on the January 20 call. This represents a fairly substantial percentage loss on his investment in a short period of time. He could do nothing, hoping for the stock to recover its loss. Unfortunately, this may prove to be wishful thinking.

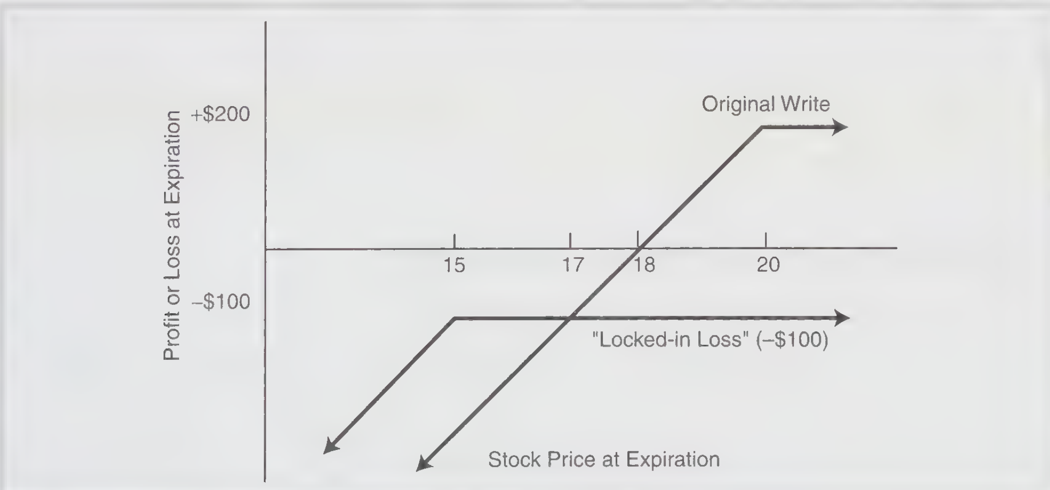
If he considers rolling down, he will not be excited by what he sees. Suppose that the writer wants to roll down from the January 20 to the January 15. He would thus buy the January 20 at .50 and sell the January 15 at 2.50, for a net credit of 2 points. By rolling down, he is obligating himself to sell his stock at 15, the striking price of the January 15 call. Suppose XYZ were above 15 in January and were called away. How would the writer do? He would lose 5 points on his stock, since he originally bought it at 20 and is selling it at 15. This 5-point loss is substantially offset by his option profits, which amount to 4 points: 1.50 points of profit on the January 20, sold at 2 and bought back at .50, plus the 2.50 points received from the sale of the January 15. However, his net result is a 1-point loss, since he lost 5 points on the stock and made only 4 points on the options. Moreover, this 1-point loss is the best that he can hope for! This is true because, as has been demonstrated several times, a covered writing position makes its maximum profit anywhere above the striking price. Thus, by rolling down to the 15 strike, he has limited the position severely, to the extent of "locking in a loss."

Even considering what has been shown about this loss, *it is still correct for this writer to roll down to the January 15.* Once the stock has fallen to 16, there is nothing anybody can do about the unrealized losses. However, if the writer rolls down, he can prevent the losses from accumulating at a faster rate. In fact, he will do better by rolling down if the stock drops further, remains unchanged, or even rises slightly. Table 2.16 and Figure 2-4

TABLE 2.16.
Profits of original write and rolled position.

Stock Price at Expiration	Profit from January 20 Write	Profit from Rolled Position
10	– $800	– $600
15	– 300	– 100
18	0	– 100
20	+ 200	– 100
25	+ 200	– 100

FIGURE 2-4.
Comparison: original write vs. "locked-in loss."



compare the original write with the rolled-down position. It is clear from the figure that the rolled-down position is locked into a loss. However, the rolled-down position still outperforms the original position unless the stock rallies back above 17 by expiration. Thus, if the stock continues to fall, if it remains unchanged, or even if it rallies less than 1 point, the rolled-down position actually outperforms the original write. It is for this reason that the writer is taking the most logical action by rolling down, even though to do so locks in a loss.

Technical analysis may be able to provide a little help for the writer faced with the dilemma of rolling down to lock in a loss or else holding onto a position that has no further downside protection. If XYZ has broken a support level or important trend line, it is added evidence for rolling down. In our example, it is difficult to imagine the case in which a $20 stock suddenly drops to become a $16 stock without substantial harm to its technical picture. Nevertheless, if the charts should show that there is support at 15.50 or 16, it may be worth the writer's while to wait and see if that support level can hold before rolling down.

Perhaps the best way to avoid having to lock in losses would be to establish positions that are less likely to become such a problem. In-the-money covered writes on higher-priced stocks that have a moderate amount of volatility will rarely force the writer to lock in a loss by rolling down. Of course, any stock, should it fall far enough and fast enough, could force the writer to lock in a loss if he has to roll down two or three times in a fairly short time span. However, the higher-priced stock has striking prices that are much closer together (in percentages); it thus presents the writer with the opportunity to utilize a new option with a lower striking price much sooner in the decline of the stock. Also, higher volatility should help in generating large enough premiums that substantial portions of the stock's decline can be hedged by rolling down. Conversely, low-priced stocks, especially nonvolatile ones, often present the most severe problems for the covered writer when they decline in price.

A related point concerning order entry can be inserted here. When one simultaneously buys one call and sells another, he is executing a spread. Spreads in general are discussed at length later. However, the covered writer should be aware that whenever he rolls his position, the order can be placed as a spread order. This will normally help the writer to obtain a better price execution.

AN ALTERNATIVE METHOD OF ROLLING DOWN

There is another alternative that the covered writer can use to attempt to gain some additional downside protection without necessarily having to lock in a loss. Basically, the writer rolls down only part of his covered writing position.

Example: One thousand shares of XYZ were bought at 20 and 10 January 20 calls were sold at 2 points each. As before, the stock falls to 16, with the following prices: XYZ January 20 call, .50; and XYZ January 15 call, 2.50. As was demonstrated in the last section, if the writer were to roll all 10 calls down from the January 20 to the January 15, he would be locking in a loss. Although there may be some justification for this action, the writer would naturally rather not have to place himself in such a position.

One can attempt to achieve some balance between added downside protection and upward profit potential by rolling down only part of the calls. In this example, the writer would buy back only 5 of the January 20's and sell 5 January 15 calls. He would then have this position:

long 1,000 XYZ at 20;
short 5 XYZ January 20's at 2;

short 5 XYZ January 15's at 2.50; and
realized gain, $750 from 5 January 20's.

This strategy is generally referred to a *partial roll-down*, in which only a portion of the original calls is rolled, as opposed to the more conventional complete roll-down. Analyzing the partially rolled position makes it clear that the writer no longer locks in a loss.

If XYZ rallies back above 20, the writer would, at expiration, sell 500 XYZ at 20 (breaking even) and 500 at 15 (losing $2,500 on this portion). He would make $1,000 from the five January 20's held until expiration, plus $1,250 from the five January 15's, plus the $750 of realized gain from the January 20's that were rolled down. This amounts to $3,000 worth of option profits and $2,500 worth of stock losses, or an overall net gain of $500, less commissions. Thus, the partial roll-down offers the writer a chance to make some profit if the stock rebounds. Obviously, the partial roll-down will not provide as much downside protection as the complete roll-down does, but it does give more protection than not rolling down at all. To see this, compare the results given in Table 2-17 if XYZ is at 15 at expiration.

TABLE 2-17.
Stock at 15 at expiration.

Strategy	Stock Loss	Option Profit	Total Loss
Original position	– $5,000	+ $2,000	– $3,000
Partial roll-down	– 5,000	+ 3,000	– 2,000
Complete roll-down	– 5,000	+ 4,000	– 1,000

In summary, the covered writer who would like to roll down, but who does not want to lock in a loss or who feels the stock may rebound somewhat before expiration, should consider rolling down only part of his position. If the stock should continue to drop, making it evident that there is little hope of a strong rebound back to the original strike, the rest of the position can then be rolled down as well.

UTILIZING DIFFERENT EXPIRATION SERIES WHEN ROLLING DOWN

In the examples thus far, the same expiration month has been used whenever rolling-down action was taken. In actual practice, the writer may often want to use a more distant expiration month when rolling down and, in some cases, he may even want to use a nearer expiration month.

The advantage of rolling down into a more distant expiration series is that more actual points of protection are received. This is a common action to take when the underlying stock has become somewhat worrisome on a technical or fundamental basis. However, since rolling down reduces the maximum profit potential—a fact that has been demonstrated several times—every roll-down should not be made to a more distant expiration series. By utilizing a longer-term call when rolling down, one is reducing his maximum profit potential for a longer period of time. Thus, the longer-term call should be used only if the writer has grown concerned over the stock's capability to hold current price levels. The partial roll-down strategy is particularly amenable to rolling down to a longer-term call since, by rolling down only part of the position, one has already left the door open for profits if the stock should rebound. Therefore, he can feel free to avail himself of the maximum protection possible in the part of his position that is rolled down.

The writer who must roll down to lock in a loss, possibly because of circumstances beyond his control, such as a sudden fall in the price of the underlying stock, may actually want to roll down to a near-term option. This allows him to make back the available time premium in the short-term call in the least time possible.

Example: A writer buys XYZ at 19 and sells a 6-month call for 2 points. Shortly thereafter, however, bad news appears concerning the common stock and XYZ falls quickly to 14. At that time, the following prices exist for the calls with the striking price 15:

XYZ common, 14;
near-term call, 1;
middle-term call, 1.50; and
far-term call, 2.

If the writer rolls down into any of these three calls, he will be locking in a loss. Therefore, the best strategy may be to roll down into the near-term call, planning to capture one point of time premium in 3 months. In this way, he will be beginning to work himself out of the loss situation by availing himself of the most potential time premium decay in the shortest period of time. When the near-term call expires 3 months from now, he can reassess the situation to decide if he wants to write another near-term call to continue taking in short-term premiums, or perhaps write a long-term call at that time.

When rolling down into the near-term call, one is attempting to return to a potentially profitable situation in the shortest period of time. By writing short-term calls one or two times, the writer will eventually be able to reduce his stock cost nearer to 15 in the shortest time period. Once his stock cost approaches 15, he can then write a long-term call with striking price 15 and return again to a potentially profitable situation. He will no longer be locked into a loss.

ACTION TO TAKE IF THE STOCK RISES

A more pleasant situation for the covered writer to encounter is the one in which the underlying stock rises in price after the covered writing position has been established. There are generally several choices available if this happens. The writer may decide to do nothing and to let his stock be called away, thereby making the return that he had hoped for when he established the position. On the other hand, if the underlying stock rises fairly quickly and the written call comes to parity, the writer may either close the position early or roll the call up. Each case is discussed.

Example: Someone establishes a covered writing position by buying a stock at 50 and selling a 6-month call for 6 points. His maximum profit potential is 6 points anywhere above 50 at expiration, and his downside break-even point is 44. Furthermore, suppose that the stock experiences a substantial rally and that it climbs to a price of 60 in a short period of time. With the stock at 60, the July 50 might be selling for 11 points and a July 60 might sell for as much as 7 points. Thus, the writer may consider buying back the call that was originally written and rolling up to the call with a higher striking price. Table 2-18 summarizes the situation.

TABLE 2-18
Comparison of original and current prices.

Original Position	Current Prices	
Buy XYZ at 50	XYZ common	60
Sell XYZ July 50 call at 6	XYZ July 50	11
	XYZ July 60	7

If the writer were to *roll-up*—that is, buy back the July 50 and sell the July 60—he would be increasing his profit potential. If XYZ were above 60 in July and were called away, he would make his option credits—6 points from the July 50 plus 7 points from the July 60—less the 11 points he paid to buy back the July 50. Thus, his option profits would amount to 2 points, which, added to the stock profit of 10 points, increases his maximum profit potential to 12 points anywhere above 60 at July expiration.

To increase his profit potential by such a large amount, the covered writer has given up some of his downside protection. *The downside break-even point is always raised by the amount of the debit required to roll up.* The debit required to roll up in this example is 4 points—buy the July 50 at 11 and sell the July 60 at 7. Thus, the break-even point is increased from the original 44 level to 48 after rolling up. There is another method of calculating the new profit potential and break-even point. In essence, the writer has raised his net stock cost to 55 by taking the realized 5-point loss on the July 50 call. Hence, he is essentially in a covered write whereby he has bought stock at 55 and has sold a July 60 call for 7. When expressed in this manner, it may be easier to see that the break-even point is 48 and the maximum profit potential, above 60, is 12 points.

Note that *when one rolls up, there is a debit incurred*. That is, the investor must deposit additional cash into the covered writing position. This was not the case in rolling down, because credits were generated. Debits are considered by many investors to be a seriously negative aspect of rolling up, and they therefore prefer never to roll up for debits. Although the debit required to roll up may not be a negative aspect to every investor, it does translate directly into the fact that the break-even point is raised and the writer is subjecting himself to a potential loss if the stock should pull back. It is often advantageous to roll to a more distant expiration when rolling up. This will reduce the debit required.

The rolled-up position has a break-even point of 48. Thus, if XYZ falls back to 48, the writer who rolled up will be left with no profit. However, if he had not rolled up, he would have made 4 points with XYZ at 48 at expiration in the original position. A further comparison can be made between the original position and the rolled-up position. The two are equal at July expiration at a stock price of 54; both have a profit of 6 points with XYZ at 54 at July expiration. Thus, although it may appear attractive to roll up, one should determine the point at which the rolled-up position and the original position will be equal at expiration. If the writer believes XYZ could be subject to a 10% correction by expiration from 60 to 54—certainly not out of the question for any stock—he should stay with his original position.

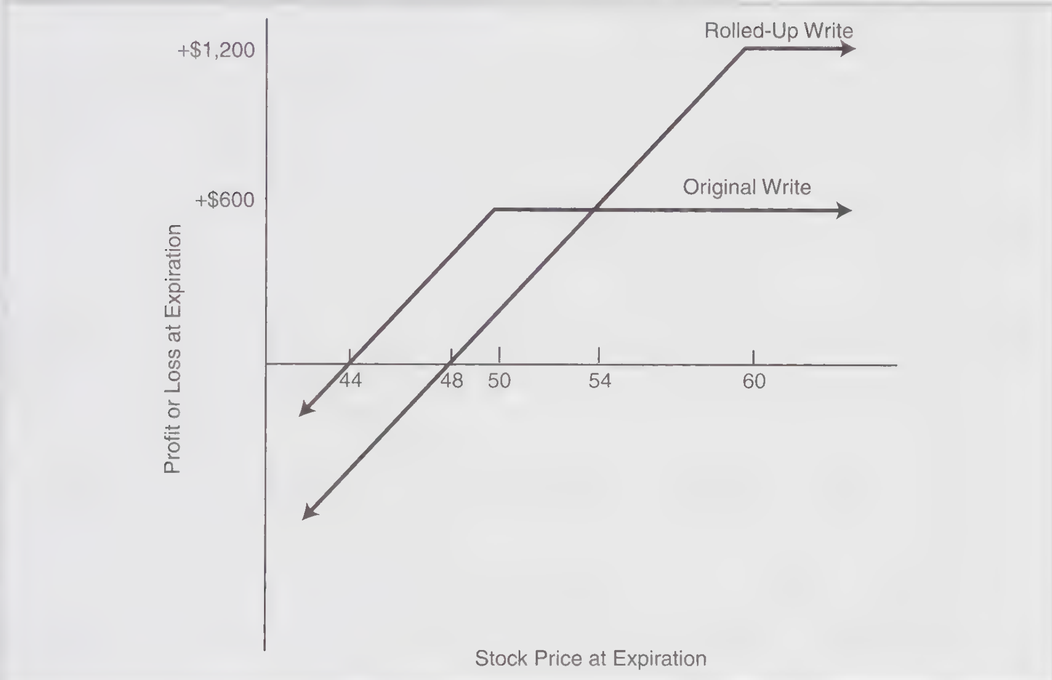
Figure 2-5 compares the original position with the rolled-up position. Note that the break-even point has moved up from 44 to 48; the maximum profit potential has increased from 6 points to 12 points; and at expiration the two writes are equal, at 54.

In summary, it can be said that rolling up increases one's profit potential but also exposes one to risk of loss if a stock price reversal should occur. Therefore, an element of risk is introduced as well as the possibility of increased rewards. Generally, it is not advisable to roll up if at least a 10% correction in the stock price cannot be withstood. One's initial goals for the covered write were set when the position was established. If the stock advances and these goals are being met, the writer should be very cautious about risking that profit.

A SERIOUS BUT ALL-TOO-COMMON MISTAKE

When an investor is overly intent on keeping his stock from being called away (perhaps he is writing calls against stock that he really has no intention of selling), then he will normally roll up and/or forward to a more distant expiration month whenever the stock rises to the strike of the written call. Most of these rolls incur a debit. If the stock is particularly strong,

FIGURE 2-5.
Comparison: original write vs. rolled-up position.



or if there is a strong bull market, these rolls for debits begin to weigh heavily on the psychology of the covered writer. Eventually, he wears down emotionally and makes a mistake. He typically takes one of two roads: (1) He buys back all of the calls for a (large) debit, leaving the entire stock holding exposed to downside movements after it has risen dramatically in price and after he has amassed a fairly large series of debits from previous rolls; or (2) he begins to sell some out-of-the-money naked puts to bring in credits to reduce the cost of continually rolling the calls up for debits. This latter action is even worse, because the entire position is now leveraged tremendously, and a sharp drop in the stock price may cause horrendous losses—perhaps enough to wipe out the entire account. As fate would have it, these mistakes are usually made when the stock is near a top in price. Any price decline after such a dramatic rise is usually a sharp and painful one.

The best way to avoid this type of potentially serious mistake is to allow the stock to be called away at some point. Then, using the funds that are released, either establish a new position in another stock or perhaps even utilize another strategy for a while. If that is not feasible, at least avoid making a radical change in strategy *after* the stock has had a particularly strong rise. Leveraging the position through naked put sales on top of rolling the calls up for debits should expressly be avoided.

The discussion to this point has been directed at rolling up *before* expiration. At or near expiration, when the time value premium has disappeared from the written call, one may have no choice but to write the next-higher striking price if he wants to retain his stock. This is discussed when we analyze action to take at or near expiration.

If the underlying stock rises, one's choices are not necessarily limited to rolling up or doing nothing. As the stock increases in price, the written call will lose its time premium and may begin to trade near parity. The writer may decide to close the position himself—perhaps well in advance of expiration—by buying back the written call and selling the stock out, hopefully near parity.

Example: A customer originally bought XYZ at 25 and sold the 6-month July 25 for 3 points—a net of 22. Now, three months later, XYZ has risen to 33 and the call is trading at 8 (parity) because it is so deeply in-the-money. At this point, the writer may want to sell the stock at 33 and buy back the call at 8, thereby realizing an effective net of 25 for the covered write, which is his maximum profit potential. This is certainly preferable to remaining in the position for three more months with no more profit potential available. The advantage of closing a parity covered write early is that one is realizing the maximum return in a shorter period than anticipated. He is thereby increasing his annualized return on the position. Although it is generally to the cash writer's advantage (margin writers read on) to take such action, there are a few additional costs involved that he would not experience if he held the position until the call expired. First, the commission for the option purchase (buy-back) is an additional expense. Second, he will be selling his stock at a higher price than the striking price, so he may pay a slightly higher commission on that trade as well. If there is a dividend left until expiration, he will not be receiving that dividend if he closes the write early. Of course, if the trade was done in a margin account, the writer will be reducing the margin interest that he had planned to pay in the position, because the debit will be erased earlier. In most cases, the increased commissions are very small and the lost dividend is not significant compared to the increase in annualized return that one can achieve by closing the position early. However, this is not always true, and one should be aware of exactly what his costs are for closing the position early.

Obviously, getting out of a covered writing position can be as difficult as establishing it. Therefore, one should place the order to close the position with his brokerage firm's option desk, to be executed as a "net" order. The same traders who facilitate establishing covered writing positions at net prices will also facilitate getting out of the positions. One would normally place the order by saying that he wanted to sell his stock and buy the option "at parity" or, in the example, at "25 net." Just as it is often necessary to be in contact with both the option and stock exchanges to *establish* a position, so is it necessary to maintain the same contacts to *remove* a position at parity.

THE PARTIAL EXTRACTION STRATEGY

This is a strategy that applies if the stock has advanced and is in danger of being called away. If the writer *wants* to be assigned, then so be it. But in many cases, the writer would like an alternative to merely buying back the call and rolling up. In this strategy, a *portion* of the underlying stock is sold, releasing proceeds that are then used to buy back the written calls. A new call option, with a higher strike, can then be written if so desired.

Example: At some time in the past, an investor wrote February 45 calls against his XYZ stock. This stock has a very low cost basis because it was purchased long ago (perhaps it was inherited). So the writer would not like to be assigned, for that would create a large tax burden. On the other hand, he is not particularly happy to pay a large debit to buy the calls back. Suppose that this is his position:

Long 3000 XYZ, currently trading at 50
Short 30 February 45 calls, currently trading at 5.00

Since the time value premium is gone from the calls, he realizes that assignment could be imminent. This lack of time value premium could be due to a large dividend payment or perhaps impending expiration—it doesn't really matter. Suppose he does the following:

Buy back 30 February 45 calls at 5.00 Cost: $15,000
Sell 300 shares of XYZ at 50 Proceeds: $15,000

Hence his net cost for this unwind of part of the covered write is zero. He is then left with 2700 shares that are "unencumbered," and he can decide if he wants to sell calls against those or not.

The advantage of this strategy is that one retains a large portion of his position, while getting out from under what he might feel is a bad situation of a call that is about to be assigned against stock that he doesn't really want to sell.

What are the tax ramifications here? First, he probably has a short-term loss on the option trade. Second, he has a long-term gain on the stock, which might be as large as most of the $15,000. But the loss on the option trade can be applied against the stock gain, so there might not be that much of a net tax exposure after all.

Moreover, this is a sort of backhanded way of selling high. After all, the stock has been rising rapidly, and it might be a good time to dispose of some shares.

Hence the practical extraction strategy has several useful features, but the major one is that it allows one to extract oneself from an in-the-money covered write with a small stock sale. It should be used before the written call gets too deeply in the money. If one waits until the option gets very deeply in-the-money before applying the technique, then a much larger number of shares will have to be sold in order to extract the stock from the covered write.

ACTION TO TAKE AT OR NEAR EXPIRATION

As expiration nears and the time value premium disappears from a written call, the covered writer may often want to *roll forward*, that is, buy back the currently written call and sell a longer-term call with the same striking price. For an in-the-money call, the optimum time to roll forward is generally when the time value premium has completely disappeared from the call. For an out-of-the-money call, the correct time to move into the more distant option series is when the return offered by the near-term option is less than the return offered by the longer-term call.

The in-the-money case is quite simple to analyze. As long as there is time premium left in the call, there is little risk of assignment, and therefore the writer is earning time premium by remaining with the original call. However, when the option begins to trade at parity or a discount, there arises a significant probability of exercise by arbitrageurs. It is at this time that the writer should roll the in-the-money call forward. For example, if XYZ were offered at 51 and the July 50 call were bid at 1, the writer should be rolling forward into the October 50 or January 50 call.

The out-of-the-money case is a little more difficult to handle, but a relatively straightforward analysis can be applied to facilitate the writer's decision. One can compute the return per day remaining in the written call and compare it to the net return per day from the longer-term call. If the longer-term call has a higher return, one should roll forward.

Example: An investor previously entered a covered writing situation in which he wrote five January 30 calls against 500 XYZ common. The following prices exist currently, 1 month before expiration:

XYZ common, 29.50;
January 30 call, .50; and
April 30 call, 2.50.

The writer can only make 50 cents more of time premium on this covered write for the time remaining until expiration. It is possible that his money could be put to better use by rolling forward to the April 30 call. Commissions for rolling forward must be subtracted from the April 30's premium to present a true comparison.

By remaining in the January 30, the writer could make, at most, $250 for the 30 days remaining until January expiration. This is a return of $8.33 per day. The commissions for rolling forward would be approximately $100, including both the buy-back and the new sale. Since the current time premium in the April 30 call is $250 per option, this would mean that the writer would stand to make 5 times $250 less the $100 in commissions during the 120-day period until April expiration; $1,150 divided by 120 days is $9.58 per day. Thus, the per-day return is higher from the April 30 than from the January 30, after commissions are included. The writer should roll forward to the April 30 at this time.

Rolling forward, since it involves a positive cash flow (that is, it is a credit transaction) simultaneously increases the writer's maximum profit potential and lowers the break-even point. In the example above, the credit for rolling forward is 2 points, so the break-even point will be lowered by 2 points and the maximum profit potential is also increased by the 2-point credit.

A simple calculator can provide one with the return-per-day calculation necessary to make the decision concerning rolling forward. The preceding analysis is only directly applicable to rolling forward at the *same striking price*. Rolling-up or rolling-down decisions at expiration, since they involve different striking prices, cannot be based solely on the differential returns in time premium values offered by the options in question.

In the earlier discussion concerning rolling up, it was mentioned that at or near expiration, one may have no choice but to write the next higher striking price if he wants to retain his stock. This does not necessarily involve a debit transaction, however. If the stock is volatile enough, one might even be able to *roll up* for even money or a slight credit at expiration. Should this occur, it would be a desirable situation and should always be taken advantage of.

Example: The following prices exist at January expiration:

XYZ, 50;
XYZ January 45 call, 5; and
XYZ July 50 call, 7.

In this case, if one had originally written the January 45 call, he could now roll up to the July 50 at expiration for a *credit* of 2 points. This action is quite prudent, since the break-even point and the maximum profit potential are enhanced. The break-even point is lowered by the 2 points of credit received from rolling up. The maximum profit potential is increased substantially—by 7 points—since the striking price is raised by 5 points and an additional 2 points of credit are taken in from the roll up. Consequently, whenever one can roll up for a credit, a situation that would normally arise only on more volatile stocks, he should do so.

Another choice that may occur at or near expiration is that of *rolling down*. The case may arise whereby one has allowed a written call to expire worthless with the stock more than a small distance below the striking price. The writer is then faced with the decision of either writing a small-premium out-of-the-money call or a larger-premium in-the-money call. Again, an example may prove to be useful.

Example: Just after the January 25 call has expired worthless,

XYZ is at 22;
XYZ July 25 call at .75; and
XYZ July 20 call at 3.50.

If the investor were now to write the July 25 call, he would be receiving only .75 of a point of downside protection. However, his maximum profit potential would be quite large if XYZ could rally to 25 by expiration. On the other hand, the July 20 at 3.50 is an attractive write that affords substantial downside protection, and its 1.50 points of time value premium are twice that offered by the July 25 call. In a purely analytic sense, one should not base his decision on what his performance has been to date, but that is a difficult axiom to apply in practice. If this investor owns XYZ at a higher price, he will almost surely opt for the July 25 call. If, however, he owns XYZ at approximately the same price, he will have no qualms about writing the July 20 call. There is no absolute rule that can be applied to all such situations, but one is usually better off writing the call that provides the best balance between return and downside protection at all times. Only if one is bullish on the underlying stock should he write the July 25 call.

AVOIDING THE UNCOVERED POSITION

There is a margin rule that the covered writer must be aware of if he is considering taking any sort of follow-up action on the day that the written call ceases trading. If another call is sold on that day, even though the written call is obviously going to expire worthless, the writer will be considered uncovered for margin purposes over the weekend and will be obligated to put forth the collateral for an uncovered option. This is usually not what the writer intends to do; being aware of this rule will eliminate unwanted margin calls. Furthermore, uncovered options may be considered unsuitable for many covered writers.

Example: A customer owns XYZ and has January 20 calls outstanding on the last day of trading of the January series (the third Friday of January; the calls actually do not expire until the following day, Saturday). If XYZ is at 15 on the last day of trading, the January 20 call will almost certainly expire worthless. However, should the writer decide to sell a longer-term call on that day without buying back the January 20, he will be considered uncovered over the weekend. Thus, *if one plans to wait for an option to expire totally worthless before writing another call, he must wait until the Monday after expiration before writing again, assuming that he wants to remain covered.* The writer should also realize that it is possible for some sort of news item to be announced between the end of trading in an option series and the actual expiration of the series. Thus, call holders might exercise because they believe the stock will jump sufficiently in price to make the exercise profitable. This has happened in the past, two of the most notable cases being IBM in January 1975 and Carrier Corp. in September 1978.

WHEN TO LET STOCK BE CALLED AWAY

Another alternative that is open to the writer as the written call approaches expiration is to let the stock be called away if it is above the striking price. In many cases, it is to the advantage of the writer to keep rolling options forward for credits, thereby retaining his stock ownership. However, in certain cases, it may be advisable to allow the stock to be called away. It should be emphasized that the writer often has a definite choice in this matter, since he can generally tell when the call is about to be exercised—when the time value premium disappears.

Example: A covered write is established by buying XYZ at 49 and selling an April 50 call for 3 points. The original break-even point was thus 46. Near expiration, suppose XYZ has risen to 56 and the April 50 is trading at 6. If the investor wants to roll forward, now is the time to do so, because the call is at parity. However, he notes that the choices are somewhat limited. Suppose the following prices exist with XYZ at 56: XYZ October 50 call, 7; and XYZ October 60 call, 2. It seems apparent that the premium levels have declined since the original writing position was established, but that is an occurrence beyond the control of the writer, who must work in the current market environment.

If the writer attempts to roll forward to the October 50, he could make at most I additional point of profit until October (the time premium in the call). This represents an extremely low rate of return, and the writer should reject this alternative since there are surely better returns available in covered writes on other securities.

On the other hand, if the writer tries to roll up and forward, it will cost 4 points to do so—6 points to buy back the April 50 less 2 points received for the October 60. This debit transaction means that his break-even point would move up from the original level of 46 to a new level of 50. If the common declines below 54, he would be eating into profits already at hand, since the October 60 provides only 2 points of protection from the current stock price of 56. If the writer is not confidently bullish on the outlook for XYZ, he should not roll up and forward.

At this point, the writer has exhausted his alternatives for rolling. His remaining choice is to let the stock be called away and to use the proceeds to establish a covered write in a new stock, one that offers a more attractive rate of return with reasonable downside protection. This choice of allowing the stock to be called away is generally the wisest strategy if both of the following criteria are met:

1. Rolling forward offers only a minimal return.
2. Rolling up and forward significantly raises the break-even point and leaves the position relatively unprotected should the stock drop in price.

SPECIAL WRITING SITUATIONS

Our discussions have pertained directly to writing against common stock. However, one may also write covered call options against convertible securities, warrants, or LEAPS. In addition, a different type of covered writing strategy—the incremental return concept—is described that has great appeal to large stockholders, both individuals and institutions.

COVERED WRITING AGAINST A CONVERTIBLE SECURITY

It may be more advantageous to buy a security that is convertible into common stock than to buy the stock itself, for covered call writing purposes. Convertible bonds and convertible preferred stocks are securities commonly used for this purpose. One advantage of using the convertible security is that it often has a higher yield than does the common stock itself.

Before describing the covered write, it may be beneficial to review the basics of convertible securities. Suppose XYZ common stock has an XYZ convertible Preferred A stock that is convertible into 1.5 shares of common. The number of shares of common that the convertible security converts into is an important piece of information that the writer must know. It can be found in a *Standard & Poor's Stock Guide* (or *Bond Guide*, in the case of convertible bonds).

The writer also needs to determine how many shares of the convertible security must be owned in order to equal 100 shares of the common stock. This is quickly determined by dividing 100 by the conversion ratio—1.5 in our XYZ example. Since 100 divided by 1.5 equals 66.666, one must own 67 shares of XYZ cv Pfd A to cover the sale of one XYZ option for 100 shares of common. Note that neither the market prices of XYZ common nor the convertible security are necessary for this computation.

When using a convertible bond, the conversion information is usually stated in a form such as, "converts into 50 shares at a price of 20." The price is irrelevant. What is important is the number of shares that the bond converts into—50 in this case. Thus, if one were using these bonds for covered writing of one call, he would need two (2,000) bonds to own the equivalent of 100 shares of stock.

Once one knows how much of the convertible security must be purchased, he can use the actual prices of the securities, and their yields, to determine whether a covered write against the common or the convertible is more attractive.

Example: The following information is known:

XYZ common, 50;
XYZ cv Pfd A, 80;
XYZ July 50 call, 5;
XYZ dividend, 1.00 per share annually; and
XYZ cv Pfd A dividend, 5.00 per share annually.

Note that, in either case, the same call—the July 50—would be written. *The use of the convertible as the underlying security does not alter the choice of which option to use.* To make the comparison of returns easier, commissions are ignored in the calculations given in Table 2.19. In reality, the commissions for the stock purchase, either common or preferred, would be very similar. Thus, from a numerical point of view, it appears to be more advantageous to write against the convertible than against the common.

When writing against a convertible security, additional considerations should be looked at. The first is the *premium of the convertible security.* In the example, with XYZ

TABLE 2.19.
Comparison of common and convertible writes.

	Write against Common	Write against Convertible
Buy underlying security	$5,000(100 XYZ)	$5,360 (67 XYZ cv Pfd A)
Sell one July 50 call	– 500	– 500
Net cash investment	$4,500	$4,860
Premium collected	$ 500	$ 500
Dividends until July	50	250
Maximum profit potential	$ 550	$ 750
Return (profit divided by investment)	12.2%	15.4%

selling at 50, the XYZ cv Pfd A has a true value of 1.5 times 50, or $75 per share. However, it is selling at 80, which represents a premium of 5 points above its computed value of 75. *Normally, one would not want to buy a convertible security if the premium is too large.* In this example, the premium appears quite reasonable. Any convertible premium greater than 15% above computed value might be considered to be too large.

Another consideration when writing against convertible securities is the *handling of assignment*. If the writer is assigned, he may either (1) convert his preferred stock into common and deliver that, or (2) sell the preferred in the market and use the proceeds to buy 100 shares of common stock in the market for delivery against the assignment notice. The second choice is usually preferable if the convertible security has any premium at all, since converting the preferred into common causes the loss of any premium in the convertible, as well as the loss of accrued interest in the case of a convertible bond.

The writer should also be aware of whether or not the convertible is *callable* and, if so, what the exact terms are. Once the convertible has been called by the company, it will no longer trade in relation to the underlying stock, but will instead trade at the call price. Thus, if the stock should climb sharply, the writer could be incurring losses on his written option without any corresponding benefit from his convertible security. Consequently, if the convertible is called, the entire position should normally be closed immediately by selling the convertible and buying the option back.

Other aspects of covered writing, such as rolling down or forward, do not change even if the option is written against a convertible security. One would take action based on the relationship of the option price and the common stock price, as usual.

WRITING AGAINST WARRANTS

It is also possible to write covered call options against warrants. Again, one must own enough warrants to convert into 100 shares of the underlying stock; generally, this would be 100 warrants. The transaction must be a cash transaction, the warrants must be paid for in full, and they have no loan value. Technically, listed warrants may be marginable, but many brokerage houses still require payment in full. There may be *an additional investment requirement*. Warrants also have an exercise price. If the exercise price of the warrant is *higher* than the striking price of the call, the covered writer must also deposit the difference between the two as part of his investment.

The advantage of using warrants is that, if they are deeply in-the-money, they may provide the cash covered writer with a higher return, since less of an investment is involved.

Example: XYZ is at 50 and there are XYZ warrants to buy the common at 25. Since the warrant is so deeply in-the-money, it will be selling for approximately $25 per warrant. XYZ pays no dividend. Thus, if the writer were considering a covered write of the XYZ July 50, he might choose to use the warrant instead of the common, since his investment, per 100 shares of common, would only be $2,500 instead of the $5,000 required to buy 100 XYZ. The potential profit would be the same in either case because no dividend is involved.

Even if the stock does pay a dividend (warrants themselves have no dividend), the writer may still be able to earn a higher return by writing against the warrant than against the common because of the smaller investment involved. This would depend, of course, on the exact size of the dividend and on how deeply the warrant is in-the-money.

Covered writing against warrants is not a frequent practice because of the small number of warrants on optionable stocks and the problems inherent in checking available returns. However, in certain circumstances, the writer may actually gain a decided advantage by writing against a deep in-the-money warrant. It is often not advisable to write against a warrant that is at- or out-of-the-money, since it can decline by a large percentage if the underlying stock drops in price, producing a high-risk position. Also, the writer's investment may increase in this case if he rolls down to an option with a striking price lower than the warrant's exercise price.

WRITING AGAINST LONG-TERM OPTIONS

A form of covered call writing can be constructed by buying LEAPS call options and selling shorter-term out-of-the-money calls against them. This strategy is much like writing calls against warrants. This strategy is discussed in more detail in Chapter 25, under the subject of diagonal spreads.

THE INCREMENTAL RETURN CONCEPT (ROLLING FOR CREDITS)

The incremental return concept of covered call writing is a way in which *the covered writer can earn the full value of stock appreciation between today's stock price and a target sale price, which may be substantially higher.* At the same time, the writer can earn an incremental, positive return from writing options.

Many institutional investors are somewhat apprehensive about covered call writing because of the upside limit that is placed on profit potential. If a call is written against a stock that subsequently declines in price, most institutional managers would *not* view this as an unfavorable situation, since they would be outperforming all managers who owned the stock and who did not write a call. However, if the stock rises substantially after the call is written, many institutional managers do not like having their profits limited by the written call. This strategy is not only for institutional money managers, although one should have a relatively substantial holding in an underlying stock to attempt the strategy—at least 500 shares and preferably 1,000 shares or more. *The incremental return concept can be* *used by anyone who is planning to hold his stock, even if it should temporarily decline in price, until it reaches a predetermined, higher price at which he is willing to sell the stock.*

The basic strategy involves, as an initial step, selecting the target price at which the writer is willing to sell his stock.

Example: A customer owns, 1,000 shares of XYZ, which is currently at 60, and is willing to sell the stock at 80. In the meantime, he would like to realize a *positive cash flow* from writing options against his stock. This positive cash flow does not necessarily result in a realized option gain until the stock is called away. Most likely, with the stock at 60, there would not be options available with a striking price of 80, so one could not write 10 July 80's, for example. This would not be an optimum strategy even if the July 80's existed, for the investor would be receiving so little in option premiums—perhaps 10 cents per call—that writing might not be worthwhile. The incremental return strategy allows this investor to achieve his objectives regardless of the existence of options with a higher striking price.

The foundation of the incremental return strategy is to write against only a part of the entire stock holding initially, and to write these calls at the striking price nearest the current stock price. Then, should the stock move up to the next higher striking price, one rolls up for a *credit* by adding to the number of calls written. Rolling for a credit is mandatory and is the key to the strategy. Eventually, the stock reaches the target price and the stock is called away, the investor sells all his stock at the target price, and in addition earns the total credits from all the option transactions.

Example: XYZ is 60, the investor owns 1,000 shares, and his target price is 80. One might begin by selling three of the longest-term calls at 60 for 7 points apiece. Table 2-20 shows how a poor case—one in which the stock climbs directly to the target price—might work. As Table 2-20 shows, if XYZ rose to 70 in one month, the three original calls would be bought back and enough calls at 70 would be sold to produce a credit—5 XYZ October 70's. If the stock continued upward to 80 in another month, the 5 calls would be bought back and the entire position—10 calls—would be written against the target price.

If XYZ remains above 80, the stock will be called away and *all 1,000 shares will be sold at the target price of 80*. In addition, the investor will earn all the option credits generated along the way. These amount to $2,800. Thus, the writer obtained the full appreciation of his stock to the target price plus an incremental, positive return from option writing.

In a flat market, the strategy is relatively easy to monitor. If a written call loses its time value premium and therefore might be subject to assignment, the writer can roll forward to a more distant expiration series, keeping the quantity of written calls constant. This transaction would generate additional credits as well.

TABLE 2.20.
Two months of incremental return strategy.

Day 1: XYZ = 60	
Sell 3 XYZ October 60's at 7	+$ 2,100 credit
One month later: XYZ = 70	
Buy back the 3 XYZ Oct 60's at 11 and	–$ 3,300 debit
sell 5 XYZ Oct 70's at 7	+$3,500 credit
Two months later: XYZ = 80	
Buy back the 5 Oct 70's at 11 and	–$ 5,500 debit
sell 10 XYZ Oct 80's at 6	+$6,000 credit
	+$2,800 credit

If the target price is eventually reached, and the writer then decides that he wants to retain some of the stock, the "partial extraction strategy" described earlier can be used when the written calls begin to lose their time value premium. Once the position is "extracted," a new, higher target can be set and the whole process begun once again.

COVERED CALL WRITING SUMMARY

This concludes the chapter on covered call writing. The strategy will be referred to later, when compared with other strategies. Here is a brief summary of the more important points that were discussed.

Covered call writing is a viable strategy because it reduces the risk of stock ownership and will make one's portfolio less volatile to short-term market movements. It should be understood, however, that covered call writing may underperform stock ownership in general because of the fact that stocks can rise great distances, while a covered write has limited upside profit potential. The choice of which call to write can make for a more aggressive or more conservative write. Writing in-the-money calls is strategically more conservative than writing out-of-the-money calls, because of the larger amount of downside protection received. The total return concept of covered call writing attempts to achieve the maximum balance between income from all sources—option premiums, stock ownership, and dividend income—and downside protection. This balance is usually realized by writing calls when the stock is near the striking price, either slightly in- or slightly out-of-the-money.

The writer should compute various returns before entering into the position: the return if exercised, the return if the stock is unchanged at expiration, and the break-even point. To truly compare various writes, returns should be annualized, and all commissions and dividends should be included in the calculations. Returns will be increased by taking larger positions in the underlying stock—500 or 1,000 shares. Also, by utilizing a brokerage firm's capability to produce "net" executions, buying the stock and selling the call at a specified net price differential, one will receive better executions and realize higher returns in the long run.

The selection of which call to write should be made on a comparison of available returns and downside protection. One can sometimes write part of his position out-of-the-money and the other part in-the-money to force a balance between return and protection that might not otherwise exist. Finally, one should not write against an underlying stock if he is bearish on the stock. The writer should be slightly bullish, or at least neutral, on the underlying stock.

Follow-up action can be as important as the selection of the initial position itself. By rolling down if the underlying stock drops, the investor can add downside protection and current income. If one is unwilling to limit his upside potential too severely, he may consider rolling down only part of his call writing position. As the written call expires, the writer should roll forward into a more distant expiration month if the stock is relatively close to the original striking price. Higher consistent returns are achieved in this manner, because one is not spending additional stock commissions by letting the stock be called away. An aggressive follow-up action can also be taken when the underlying stock rises in price: The writer can roll up to a higher striking price. This action increases the maximum profit potential but also exposes the position to loss if the stock should subsequently decline. One would want to take no follow-up action and let his stock be called if it is above the striking price and if there are better returns available elsewhere in other securities.

Covered call writing can also be done against convertible securities—bonds or preferred stocks. These convertibles sometimes offer higher dividend yields and therefore increase the overall return from covered writing. Also, the use of warrants in place of the underlying stock may be advantageous in certain circumstances, because the net investment is lowered while the profit potential remains the same. Therefore, the overall return could be higher.

Finally, the larger individual stockholder or institutional investor who wants to achieve a certain price for his stock holdings should operate his covered writing strategy under the incremental return concept. This will allow him to realize the full profit potential of his underlying stock, up to the target sale price, and to earn additional positive income from option writing.